Reference Pricing as a Focal Point: Evidence from the Finnish Pharmaceutical Market *

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Abstract

While price regulation is widely used to decrease prices, it can inadvertently provide firms with a focal point for coordination. The reference price system adopted in the Finnish pharmaceutical market in 2009 initially resulted in lower prices, but after a few years, prices gradually returned to their pre-reform level. How did firms succeed in raising prices, and what role did the regulation play in this process? We find that firms learned to use the previous quarter's reference price as a focal point when choosing prices at the beginning of each quarter. This pattern led to price increases from quarter to quarter. Six big generic firms were more likely to use focal-point pricing, and their presence was associated with similar behavior among smaller rivals. According to our structural model, 65% of products had a profitable downward deviation from the upward pricing path but only 13% deviated. The remaining 52% suggests potential tacit coordination. On the positive side, we show that after the quarterly reference price was fixed at the beginning of the quarter, it constrained firms from increasing prices higher.

JEL codes: L51, L13, L11, L65, I18

Keywords: Price regulation, tacit coordination, pharmaceutical markets

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1 Introduction

While price regulation is widely used to decrease prices, it can inadvertently provide firms with a publicly observable focal point for coordination, as documented in markets such as credit cards, grocery, and gasoline (Knittel and Stango, 2003; Genakos et al., 2018; Martin and Verboven, 2026). The Finnish pharmaceutical reference price system provides a compelling institutional setting to study how price regulation can both restrain prices and facilitate price coordination. Introduced in 2009, the system initially resulted in lower prices, but after a few years, prices gradually returned to their pre-reform level. How did firms succeed in raising prices, and what role did the regulation play in this process?

Our study of the Finnish pharmaceutical market provides lessons not only for other pharmaceutical markets but also for price regulation in general. Navigating price regulation is a learning process for both firms and the regulator. In response to the increase in prices, the Finnish regulator made a minor change to the system in 2017 and succeeded in reversing the price trend. Why did the change in 2017 result in lower prices?

Using product-level wholesale data covering almost 1500 markets from 2009 to 2022, our descriptive analysis documents four stylized facts. First, up to 60% of products had prices equal to the previous quarter's reference price at the beginning of the quarter, which we refer to as *focal-point pricing*. The reference price for the whole quarter was determined by the lowest price at the beginning of the quarter plus a regulated price band. This regulation could facilitate reference-price increases through the following mechanism: At the beginning of the quarter, if all firms in a market chose prices equal to the previous quarter's reference price, the new reference price increased by exactly the price band. By repeatedly using focal-point pricing, firms could generate stepwise increases in the reference price from quarter to quarter. We observe such *price-band-sized reference-price increases* in up to 30% of markets.

Second, firms could change their prices every two weeks, but the reference price was fixed for the whole quarter. Firms had a strong incentive to keep their prices at or below the reference price because pharmacies recommended only products priced at or below it, and consumers were not reimbursed for the price difference above the reference price. Indeed, after the quarterly reference price was fixed at the beginning of the quarter, only 7% of products were priced above it, and 50% of products were priced exactly at it.

Third, six big generic firms, which we collectively refer to as the *Big Six*, were considerably more likely to use focal-point pricing than the other firms in the market. Our descriptive analysis shows that the Big Six differed substantially from the other firms in terms of sales, the number of products supplied, the timing of price changes, and the tendency to undercut the previous reference price. The reference price was more likely to increase in markets

containing only Big Six firms. Also, in markets containing both Big Six and the other firms, a larger share of products supplied by the Big Six was positively correlated with the likelihood of a reference-price increase.

Fourth, alarmed by the increase in prices, the regulator decided to change the price band when calculating the reference price with the objective of lowering prices. Between 2009 and 2016, the price band was €1.5 or €2. After 2017, the regulator changed the price band to €0.5. Following the 2017 reform, we observe declines in both focal-point pricing at the product level and price-band-sized reference-price increases at the market level. The share of products with focal-point pricing decreased from 60% in 2013-2016 to 35% after 2017. Similarly, the share of markets with price-band-sized reference-price increases dropped from 23-30% in 2013-2016 to 10-15% after 2017.

Building on these four stylized facts, we construct a structural model to study firms' pricing incentives and whether observed prices can be explained by competitive pricing or potential coordination. We estimate a discrete-choice demand model that captures substitution across similar products within a market. Consumers choose a product in the market based on prices, whether a product is priced at or below the reference price, and whether it is generic, branded, or parallel-imported. Our market definition follows the regulator's definition of a reference price group, which includes products with the same active ingredient, strength, dosage form, and package-size group. For example, a market may contain all 10-mg 30-pill paracetamol products. Our demand estimates confirm that products priced at or below the reference price and branded products have higher market shares. Demand is more price elastic for products priced at or below the reference price than for products priced above it.

On the supply side, following the related approach of using observed prices as marginal costs in Tunçel (2025) and Dubois et al. (2022), we construct marginal costs using each product's minimum observed price over the sample period, adjusted over time using a monthly producer price index. We do not recover marginal costs from firms' first-order conditions for two reasons. The first reason is the presence of corner solutions: Products priced at the reference price have substantially higher sales than products priced just one cent above it. Second, we are unwilling to impose an assumption on firms' pricing behavior as we want to analyze whether prices were competitive or potentially coordinated. To model firms' pricing decisions, we distinguish between the beginning of the quarter, when the reference price is set, and the remainder of the quarter, after the reference price is fixed.

Our structural model provides three main findings. First, we begin by analyzing firms' incentives during the reference-price-setting period at the beginning of each quarter and find that 52% of observed prices are consistent with potential coordination. We calculate

whether each product had an incentive to unilaterally deviate downward by choosing a sufficiently low price to be the only product eligible for the full reimbursement and pharmacy recommendation at the beginning of the quarter. If a firm did not deviate despite having a profitable downward deviation, we interpret the observation as potential coordination. Analyzing firms' incentives at the beginning of the quarter, our structural model suggests that 65% of products had a profitable downward deviation from the upward pricing path, but only 13% deviated. The remaining 52% suggests potential coordination.

Second, we study whether the reference price was effective in restraining prices during the fixed-reference-price periods. For products priced exactly at the reference price during these periods, we calculate whether they would have unilaterally increased their prices if they had retained their reimbursement and pharmacy recommendation status. We find that 83% of products observed at the reference price would have unilaterally increased their prices in the absence of the reference price. This result implies that, during the fixed-reference-price periods, 83% of prices at the reference price can be rationalized by competitive pricing subject to the reference-price constraint. To obtain a lower-bound estimate of how often the fixed reference price restrained firms from pricing higher, we repeat the exercise assuming zero marginal costs and obtain an estimate of 29%.

Third, after the narrower price band was introduced in 2017, firms deviated more often. After 2017, 23% of products with profitable deviations actually deviated, compared to 15% in 2013-2016. Our structural model offers an intuitive explanation for the effectiveness of the narrower price band: It was easier to push rivals above the reference price (and get the exclusive reimbursement and pharmacy recommendation status) by undercutting them when the price band was €0.5 than when it was €1.5 or €2.

Taken together, our results suggest that, although the reference price system facilitated focal-point pricing and reference-price increases at the beginning of the quarter, the fixed reference price prevented many firms from increasing prices higher within the quarter. In addition, after firms learned to game the system from 2009 to 2017, the regulator learned to adjust the regulation in response. The narrower price band introduced in 2017 increased price competition by reducing firms' incentives to use focal-point pricing.

Our first contribution is to the empirical literature on focal-point pricing and tacit coordination. While Knittel and Stango (2003) show how state-level interest-rate ceilings in the U.S. credit card market create a focal point for issuers, Genakos et al. (2018) provide evidence that the regulated maximum markup in the Greek fruit and vegetable market acts as a focal point for coordination. More recently, Martin and Verboven (2026) find that price caps in the Belgian retail gasoline market constrain only a minority of firms while inducing many firms to coordinate at the cap or at integer discounts below it. In these studies of

regulation-induced focal points, the focal point is externally set by the regulator. To our knowledge, we are the first to study how regulation can generate an endogenous focal point that evolves with firms' past pricing decisions.

Also studying focal-point pricing, Byrne and De Roos (2019) document a gradual transition to a coordinated equilibrium in the retail gasoline market in Perth, Australia in which dominant firms use price leadership to create focal points and help rivals learn to coordinate. Similar to their approach, we begin by documenting heterogeneity in pricing behavior across firms. In particular, we show that six big generic firms were substantially more likely than the other firms to use focal-point pricing at the beginning of the quarter. The difference of our setting is that firms did not need a dominant firm to create and signal the focal price, the regulation itself did it.

We also contribute to the literature on reference price regulation in pharmaceutical markets. Kortelainen et al. (2023) find that the Finnish reference price system decreased the expenditure per dose by 13% in the three years following its introduction in 2009. Other studies of pharmaceutical reference price systems also demonstrate that they strengthen price competition and reduce expenditures in Denmark (Kaiser et al., 2014), Norway (Brekke et al., 2009, 2011), and Sweden (Granlund and Bergman, 2018). Our paper complements this literature by showing that a reference price system could facilitate price increases, and the regulator needs to learn to adapt to firms' pricing behavior.

Finally, our paper contributes to the literature on pricing patterns and collusion in pharmaceutical markets. Previous studies document cartels in the US pharmaceutical market (Clark et al., 2022; Starc and Wollmann, 2025), recurring price cycles in the Danish pharmaceutical market (Hauschultz and Munk-Nielsen, 2020), price dynamics (Janssen, 2022), and collusion through entry decisions in different package sizes (Le and Toivanen, 2026) in the Swedish pharmaceutical market. To the best of our knowledge, our paper is the first to identify the two sides of reference pricing: restraining prices on the one hand, and providing a focal point for price coordination on the other hand.

The remainder of the paper is organized as follows. In Section 2, we describe the regulations and our data. In Section 3, we document the aggregate development of reference prices and focal-point pricing over time. In Section 4, we outline the differences between the Big Six and the other firms. In Section 5, we present a descriptive analysis of focal-point pricing. In Section 6, we develop and estimate our structural model, and use it to analyze firms' pricing decisions. In Section 7, we conclude.

2 Regulations and Data

In this section, we describe the prescription practice, the individual- and product-specific reimbursement rate, the mechanism for setting the reference price, the price band, the regulated maximum price cap, and our datasets.

2.1 Prescribing and Reimbursement

In Finland, physicians prescribe pharmaceutical products by specifying the active ingredient, strength, dosage form, quantity required, and, in some cases, the firm name. Since the introduction of generic substitution in Finland in April 2003, pharmacists have been required to offer patients the cheapest or nearly cheapest generic medicine suitable for substitution in place of the prescribed product. Before 2009, consumers' incentives to accept pharmacists' offer was low because they faced the same reimbursement rule regardless of whether they accepted or refused the offer (Kortelainen et al., 2023).

The reimbursement rate for prescription pharmaceuticals depends on the illness severity, the product's active ingredient and the consumer's annual out-of-pocket pharmaceutical expenditure. Consumers have to pay out-of-pocket up to an initial annual deductible (€50 in 2016-2022) before receiving reimbursement. Depending on the illness severity and the product's active ingredient, the reimbursement rate ranges from 40% to 100%. After reaching the annual ceiling on out-of-pocket pharmaceutical expenditure (between €600 and €700), consumers only need to pay a fixed fee of €2.50 for each reimbursable product.

In April 2009, the financial incentives for consumers to accept generic substitution were strengthened through the introduction of the reference price system. Pharmacies must offer a cheaper similar product priced at or below the reference price if the prescribed product's price exceeds the reference price. If a consumer refuses to switch, they must pay the difference between the product's retail price and the reference price entirely out of pocket, unless the prescribing physician has prohibited substitution. This additional payment does not count toward the annual deductible or the out-of-pocket pharmaceutical expenditure used to determine reimbursement rates. Consumers get the same reimbursement rate for all products priced at or below the reference price.

2.2 Price Regulations

Pharmaceutical companies set uniform nationwide wholesale price on the first and fifteenth day of each month. Prices must be submitted in advance, and firms do not observe competitors' forthcoming pricing decisions before all prices are published simultaneously. Pharmacy

markups are regulated by law and are weakly increasing in wholesale prices. More details are in Table **??**. Retail prices also include the value-added tax (VAT) of 8-14%.

The regulator, Pharmaceuticals Pricing Board Hila, calculates a reference price for each reference price group on a quarterly basis. A reference price group consists of products with the same active ingredient, strength, dosage form, and package size group. The reference price is the lowest price among available products at the beginning of the quarter, plus the applicable price band. Because firms do not know rivals' prices in advance, the reference-price-setting mechanism resembles a first-price sealed-bid auction. Between 2009 and 2016, the price band was €1.5, or €2 when the VAT-inclusive price of the cheapest product exceeded €40. Since 2017, a uniform price band of €0.5 has applied to all reference price groups. Before the change in the price band in 2017, a government proposal acknowledged that prices had been increasing in some markets, and the main goal in narrowing the price band was to increase price competition (Government of Finland, 2016), but the proposal did not identify focal-point pricing as we document in this paper.

The reason for having the price band and fixing the reference price for the whole quarter is to maintain demand predictability for pharmaceutical firms, wholesalers, and pharmacies. Without sufficient predictability, manufacturing, importing, and managing inventories would be more difficult (Government of Finland, 2002).

In addition to the reference price, regulations include maximum price caps. Products are excluded from the reimbursement system if their prices exceed the maximum price caps. The maximum price caps for branded products are determined through negotiations between branded pharmaceutical firms and the regulator. For generic products, the maximum price caps are generally set as a percentage of the maximum price caps of the equivalent branded product. The percentage was 60% in 2009-2016 and 50% in 2016-2022.

2.3 Data

Our first main dataset is monthly product-level data on wholesale sales and prices from April 2009 to December 2020 from the Finnish Medicines Agency (Fimea). Our second main dataset is semimonthly product-level price data from April 2009 to January 2022 from the Association of Finnish Pharmacists (AFP). Both datasets include the Nordic Article Number (product id), market authorization holder (firm name), active ingredient, strength, package size, dosage form, and reference price group.

We use the semimonthly dataset for the descriptive analysis of prices. We use the monthly dataset for our structural model and analyses that require both price and sales data.[1] We

[1] We are in the process of acquiring the semimonthly data on sales and prices for all analyses.

complement this data with monthly product-level wholesale data on sales and prices in Norway from April 2009 to October 2018 from Farmastat for our demand estimation and event studies (reported in the Appendix **??**).

We also use data from the Finnish Pharmaceuticals Pricing Board (Hila) to exclude products with reported availability issues at the beginning of the quarter (12% of all observations) and reference price groups that are formed without generic competition.

3 The Development of Reference Prices and Focal-point Pricing

3.1 The Development of Reference Prices

We start by analyzing the development of the average reference price from the introduction of the reference price system in 2009 until 2020. In Figure 1, we plot the sales-weighted average reference price. The sales used for weighting are from 2016Q4 before the price band was reduced to €0.5 in 2017.[2,3]

Reference prices dropped 40% by 2010Q2 following the introduction of the reference price system in April 2009, consistent with previous literature showing that the 2009 reform reduced average expenditure per dose in the Finnish pharmaceutical market in the 36 months after its introduction (Kortelainen et al., 2023). Between 2011 and 2016, reference prices gradually increased to levels above those observed in 2009. As we document below (Figure 3), this increase coincided with firms' frequent use of the previous quarter's reference price as a focal point.

In 2017, the price band was reduced from €1.5/€2 to €0.5, after which reference prices began to decline. This decline coincided with a reduction in firms' use of focal-point pricing (Figure 3 below). Our objective is to analyze how focal-point pricing generated this pattern in reference prices.

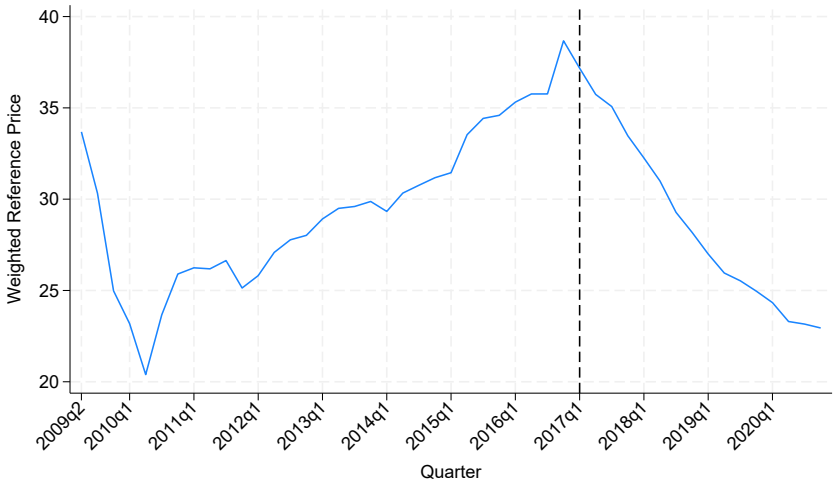
3.2 Illustrative Market of Focal-point Pricing

We illustrate the tacit coordination mechanism with an example market. Figure 2 plots two-week prices in this example market from 2014 to 2016. At the beginning of each quarter of

[2]In July 2016, the regulator also lowered the price caps of some branded products to the level applying to generic products. However, affected branded products at the price cap do not determine the minimum price in their reference price groups and therefore do not affect reference prices.

[3]As a robustness check, Figure A-1 in the Appendix presents the average reference price weighted by the number of sold packages in 2016Q4. The resulting pattern is qualitatively similar.

Figure 1: Average Reference Price



Notes: Average reference price weighted by sales in 2016Q4. Data source: AFP (2009—2022), Hila (2009–2022), Fimea (2009–2020).

the year (two-week periods 1, 7, 13, etc. in Figure 2), all three firms in the example market frequently submitted prices exactly equal to the previous quarter's reference price. As a result, the previous quarter's reference price became the minimum price at the beginning of the new quarter. Since the new reference price was calculated by adding the price band (€1.5 in this market) to the minimum price, the reference price increased mechanically by €1.5. Once the reference price was fixed for the entire quarter, all firms increased their prices by €1.5 to match it. They then kept their prices unchanged for the remainder of the quarter and submitted the same prices at the beginning of the next quarter. Over two years, prices in this market rose from €20 to €32, i.e., by 60%.[4]

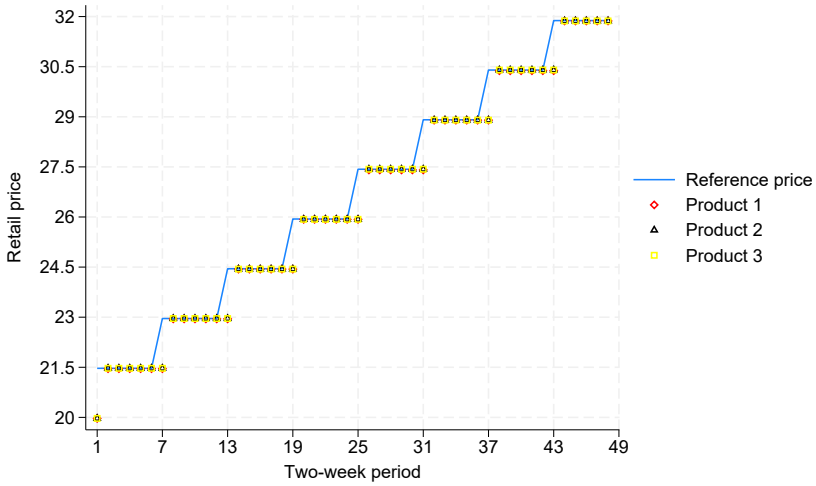
While this market nicely illustrates how the reference price system offers a focal point for firms to coordinate, it is an extreme example from the data. We next turn to a more thorough descriptive analysis of focal-point pricing.

3.3 The Development of Focal-point Pricing

We next document that pricing at the previous quarter's reference price is a systematic feature of the Finnish pharmaceutical market rather than an isolated phenomenon in one reference price group. Figure 3 focuses on prices at the beginning of each quarter and shows that a substantial share of products were priced exactly at the previous quarter's reference

[4]Our data for this market cover a longer period than the two years shown in Figure 2. In Appendix Figure A-2, we plot the reference price and the number of firms for this reference price group from 2009 to 2020. The reference price follows the aggregate pattern shown in Figure 1.

Figure 2: Prices in a Market

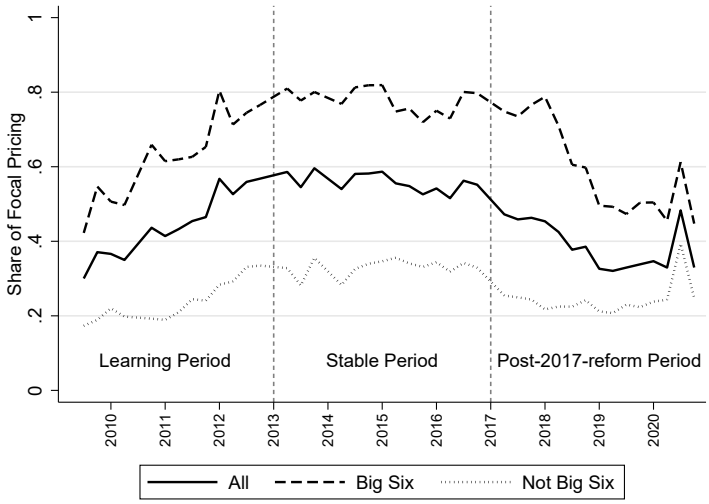


Notes: Two-week prices in an example market. Periods 1, 7, 13, 19, etc. are the beginning of the quarter. Data source: AFP (2014-2016).

price. We report the share of products using focal-point pricing with the solid line. This share rose sharply from around 30% in 2009 to roughly 60% by the end of 2012. We refer to this period as the *Learning Period*. The share then remained relatively stable at around 60% between 2013 and 2016, which we call the *Stable Period*. The Stable Period coincides with the upward trend in reference prices shown in Figure 1. Following the 2017 narrowing of the price band to €0.5, the share declined from around 60% to approximately 35%, which is close to its level in 2009. We refer to the period with the narrower price band as the *Post-2017-reform Period*. The weakening of focal-point pricing after the narrowing of the price band also coincides with the decline in reference prices shown in Figure 1.

The prevalence of focal-point pricing varies substantially across firms, as shown in Figure 3. In particular, the dashed line shows that products supplied by six big generic firms (the "Big Six") were considerably more likely to use focal-point pricing than products supplied by the other firms, shown by the dotted line. In Figure 3, the share of Big Six's products using focal-point pricing (the dashed line) doubled from around 40% in 2009 to around 80% during the Stable Period before declining back to around 40% by 2020. The share of products from the other firms engaging in focal-point pricing (the dotted line) followed the same pattern but to a lesser extent, increasing from 20% in 2009 to 35% during the Stable Period and then declining back to 20% after 2017. In the next Section, we provide more evidence that the Big Six were substantially different from the other firms.

Figure 3: Share of Products using Focal-point Pricing



Notes: Quarterly share of products priced at or within 1 cent from the previous quarter's reference price at the beginning of the quarter. Unit of observations: Product-by-quarter. Products whose prices are always at the price caps and monopoly products are excluded. Data source: Fimea (2009–2020).

4 The Big Six

We call a group of six big generic firms the "Big Six". Figure 3 in the previous section shows that these firms were considerably more likely than the other firms to use focal-point pricing at the beginning of the quarter by pricing at the previous reference price. The following subsections document three features that distinguish the Big Six from the other firms in the market. First, we show that the Big Six were among the largest firms in the Finnish reference price system in terms of both sales and the number of products supplied. Second, we examine the timing of price changes and show that the Big Six were less likely to change their prices at the turn of each quarter (when prices determine the new reference price). Finally, we examine prices at the beginning of the quarter and show that the Big Six were less likely to undercut the previous reference price substantially in order to reduce the reference price.

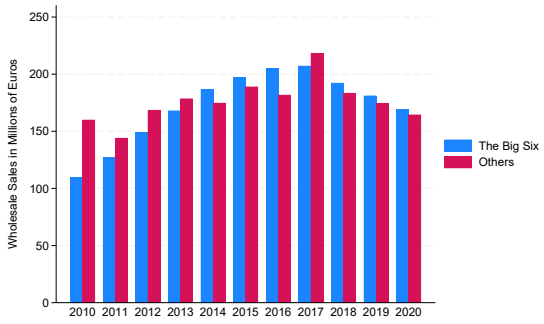
Although branded products account for 30% of the market, we do not include them in the Big Six because their influence on reference prices is limited. The reference price is determined by the minimum price in each market. Branded products' prices are often at the regulated maximum price cap (55-75% of the time). Conditional on not being at the maximum price cap, branded products are the minimum-priced product in only 2-11% of observations. More details on branded products can be found in Appendix Section A-4.

4.1 Sales and Number of Products

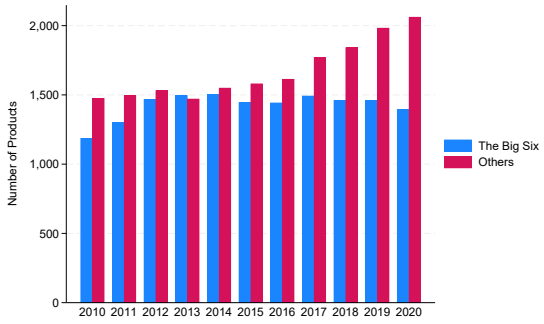
The Big Six constitute an economically important part of the Finnish reference price system. Between 2011 and 2016, products supplied by the Big Six accounted for roughly half of total pharmaceutical sales in the reference price system, as shown in Figure 4a. Over the same period, the number of unique products supplied by the Big Six was approximately equal to the number supplied by all other firms combined, as shown in Figure 4b.

Figure 4: Annual Sales and Number of Products

a) Sales in Nominal Millions of Euros



b) Number of Products



Notes: Figure a: Annual sales in wholesale prices. Figure b: Annual number of unique product numbers. Data Source: AFP (2010-2020), Fimea (2010-2020), Hila (2010-2020).

The distinction between the two groups is also apparent at the firm level. Table 1 shows that the Big Six firms were substantially larger than the other firms in 2013. A Big Six firm had, on average, 221 products and annual sales of 28 million euros, compared with 18 products and annual sales of 2.3 million euros for the average firm among the other firms (a group of 76 firms). The difference is also evident in the medians: The median Big Six firm had 224 products and annual sales of 15.8 million euros, whereas the corresponding values among the other firms were only 6 products and 0.8 million euros.

Table 1: Firm-Level Summary Statistics by Firm Group in 2013

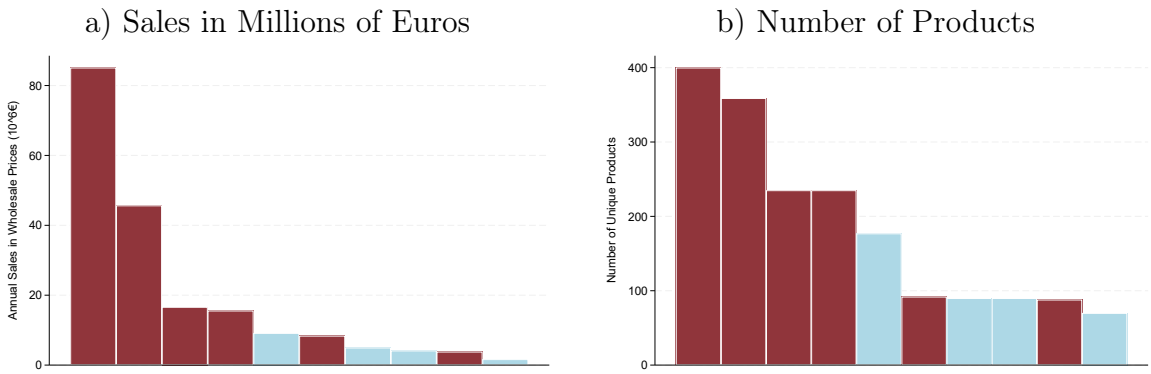
Firm group	No. of Firms	No. of Products		Sales (Millions €)	
		Mean	Median	Mean	Median
Big Six	6	221	224	28.0	15.8
Other firms	76	18	6	2.3	0.8

Notes: Data Source: AFP (2013), Fimea (2013).

Figure 5 further illustrates the importance of the Big Six by ranking the ten largest non-

branded firms in 2013 by sales and number of products.[5] The four largest firms under both measures belonged to the Big Six (highlighted by the maroon bars). We also include the two smaller firms in the Big Six because each was subsequently merged with one of these largest four.[6]

Figure 5: Firm-level Annual Sales and Number of Products in 2013 - Top 10



Notes: Maroon bars: Big Six, blue bars: the other firms. Figure a: Annual sales in wholesale prices. Figure b: Annual number of unique product numbers. Data Source: AFP (2013), Fimea (2013).

4.2 Timing of Price Changes

We illustrate how the pricing behavior of the Big Six differed from that of the other firms in the timing and direction of price changes in Figure 6. The reference price was determined at the beginning of each quarter and remained fixed for the rest of the quarter, whereas firms could revise their prices twice per month. We therefore divide each quarter into six two-week periods. Only price changes made in the first two-week period affected the reference price for a given quarter.

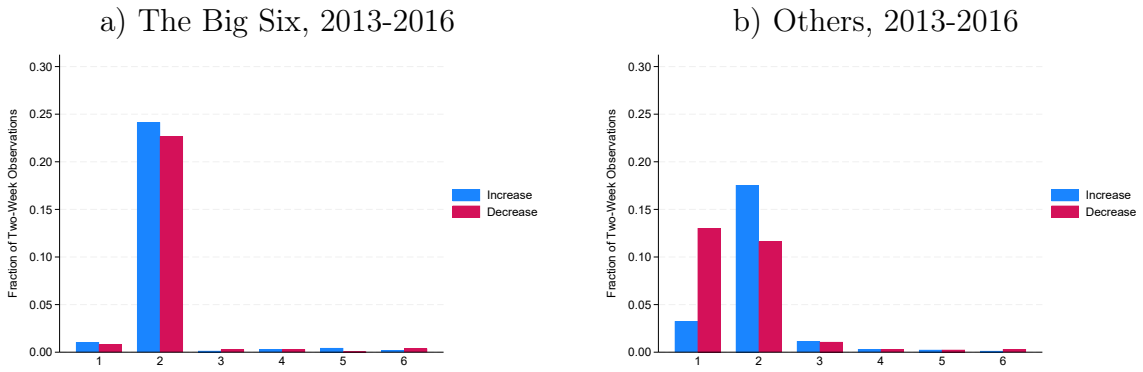
Figure 6 reports separately the fraction of products with a price increase and the fraction with a price decrease in each of the six two-week periods within a quarter in the Stable Period of 2013-2016. Bar cluster "1" captures price changes in the first two-week period of the current quarter, when prices determined the new reference price. At the beginning of the quarter, firms outside the Big Six were substantially more likely to reduce their prices in the first two-week period (13% of the other firms compared to 1% of the Big Six, red bars

[5]To verify that the Big Six were already among the major firms when the reference price system was introduced, we conduct the same exercise for 2009 in Figure A-3 in the Appendix, ranking the ten largest market authorization holders by generic sales and by the number of products. The results show that the Big Six were also among the largest firms at the beginning of the reference price regulation in 2009.

[6]The Big Six consist of six firms with different company names. In terms of ownership, the number of independent firms decreased from six to three because of mergers and acquisitions.

of bar clusters "1" in the right-hand side and left-hand side graphs respectively), when price changes could affect the new reference price.

Figure 6: Distribution of the Timing of Price Changes



Notes: x-axis: Two-week periods within a quarter. Data source: AFP (2013–2016).

Bar cluster "2" captures price changes in the second two-week periods, after the new reference price was fixed. For both the Big Six and the other firms, price changes were concentrated in the second two-week period of the quarter. The total share of products with price changes in the second two-week period (sum of red and blue bars in bar clusters "2") is 47% for the Big Six and 29% for the other firms. Many firms adjusted their prices only after observing the newly established reference price. Conditional on a price change in the second two-week period, the Big Six revised prices to the reference price 99% of the time.[7]

Other bar clusters from "3" to "6" show that all firms rarely changed their prices in the second and third month of each quarter. Figure 6 suggests that the other firms were notably more likely to decrease their prices at the beginning of the quarter, whereas the Big Six almost never adjusted their prices at the beginning of the quarter. Big Six firms waited until the new reference price was fixed before adjusting their prices in the second two-week period.

To verify the robustness of our Big Six classification, Appendix Figure A-4 shows that the timing of price changes among the non-Big-Six firms in the top 10 largest firms (in Figure 5) follows a pattern more similar to the other firms in Figure 6 than to the Big Six.

4.3 Prices at the Beginning of the Quarter

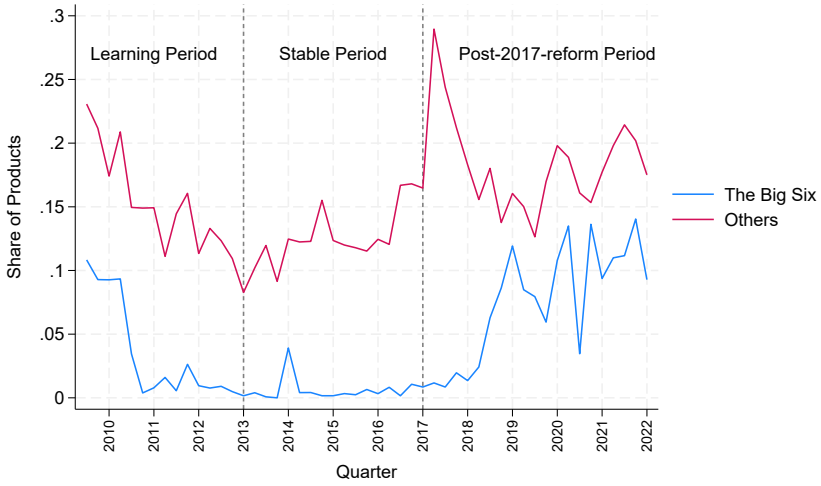
In the previous subsection, we showed that the other firms were more likely to decrease prices than the Big Six at the beginning of the quarter. Did the other firms try to decrease the

[7]Further details in Figures A-6 and A-7.

minimum price (and hence the reference price)? In this subsection, we examine the share of products whose price at the beginning of the quarter was set low enough to reduce the reference price.

In Figure 7, the red line represents the share of products with reference-price-decreasing prices for the other firms, and the blue line represents this share for the Big Six. The red line is always higher than the blue line, which means the other firms were more likely to decrease the reference price than the Big Six. In the Learning Period of 2009-2012, the shares declined for both groups of firms. In the Stable Period of 2013-2016, the blue line is close to 0, indicating that the Big Six almost never chose a price that would reduce the reference price whereas the red line indicates that for the other firms, 10-16% of products had reference-price-decreasing prices. In the Post-2017-reform period of the narrower price band, both lines increased and the gap between them narrowed.

Figure 7: Share of Products with reference-price-decreasing Prices



Notes: Share of prices at the beginning of the quarter that would decrease the reference price. Data source: AFP (2009–2022), Hila (2009-2022).

Figure 7 shows that the other firms were consistently more likely to price sufficiently below the previous reference price to reduce the reference price for the new quarter. The Big Six, by contrast, rarely made such bids. To verify the robustness of our Big Six classification, Appendix Figure A-5 shows that the share of products with reference-price-decreasing prices among the non-Big-Six firms in the top 10 largest firms (in Figure 5) follows a pattern more similar to the other firms in Figure 7 than to the Big Six.

Together with Figure 6 displaying the timing and direction of price changes, Figure

7 reveals a distinctive pricing pattern among the Big Six in the Stable Period of 2013-2016. These firms typically revised their prices in the second two-week period, after the new reference price was fixed. Conditional on a price change in the second two-week period, they revised prices to the reference price 99% of the time. Then, they rarely made price changes during the remainder of the quarter. At the beginning of the next quarter, they were also very unlikely to revise prices and frequently submitted prices at the previous reference price. The pricing behavior of the other firms was different. These firms were more likely to change prices at the beginning of the quarter, when their prices could affect the reference price. Also, the other firms were more likely to undercut the previous reference price substantially to reduce the reference price.

5 Descriptive Analysis of Focal-point Pricing

In Figure 3 in Section 3, we showed at the product level that the share of products using focal-point pricing increased from 30% in 2009 to 60% in 2012 and remained stable until 2017 before dropping to 35% after the change in the price band. In this section, we analyze market-level outcomes related to focal-point pricing, including reference-price increases, markets reaching the maximum price cap, and the number of firms in the market. We also use descriptive regressions to examine which markets are statistically more likely to experience a reference-price increase.

5.1 Price-band-sized Reference-price Increases

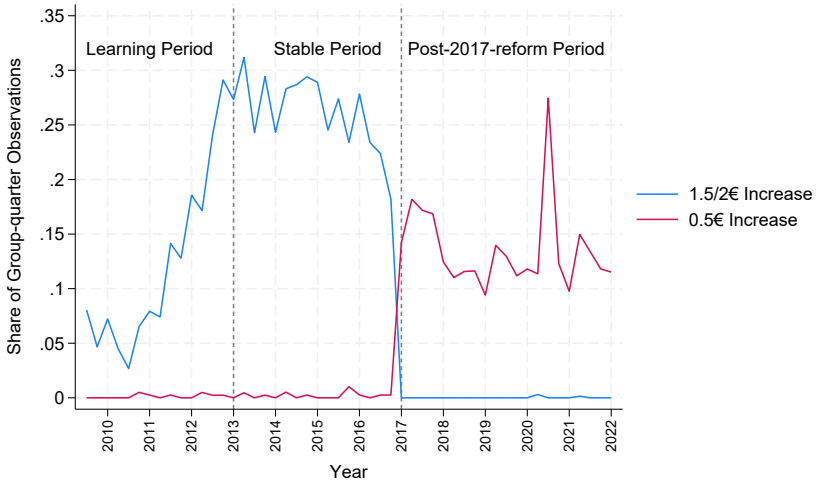
We first report the share of markets in which firms succeed in raising the reference price by exactly the price band (conditional on the maximum price cap of the market not binding), which we refer to as *price-band-sized reference-price increases*. For the focal-point pricing coordination to succeed, all firms in the market must use focal-point pricing at the beginning of the quarter.

Figure 8 reports the share of markets in which the reference price increased by exactly €1.5, €2, or €0.5. The blue line corresponds to reference-price increases of €1.5 or €2 (the pre-2017 price bands), while the red line corresponds to reference-price increases of €0.5 (the post-2017 price band).

In the Learning Period of 2009-2012, the share of markets with reference-price increases of €1.5 or €2 rose from approximately 5% in 2009 to 25% in 2012. In the Stable Period of 2013-2016, 24%-30% of markets experienced a reference-price increase equal to the applicable price band of €1.5 or €2. In these first two periods, the share of markets with reference-price

increases of €0.5 was close to 0. In contrast, in the Post-2017-reform Period of the narrower price band, the share of markets with reference-price increases of €0.5 was 10%-15%. In this last period, almost no markets had reference-price increases of €1.5 or €2. Figure 8 indicates that firms adjusted focal-point pricing to the new price band after 2017, and that the applicable price band played an important role in shaping market-level reference-price increases.

Figure 8: Price-Band-Sized Reference-price Increases



Notes: Annual share of reference-price-group-by-quarter with reference-price increases equal to the price band or within 2 cents smaller. Markets with binding price caps are excluded. Data source: AFP (2009–2022).

5.2 Reference Price Reaching the Maximum Price Cap

Pharmaceutical firms do not price above the maximum price caps set by the regulator because the products priced above the maximum price cap are excluded from the reimbursement system. If all firms in the market repeatedly practice focal-point pricing, the reference price increases quarter after quarter until it reaches the maximum price cap. In this subsection, we study the likelihood of a market reaching its maximum price cap.

In order to analyze whether a market reaches the maximum price cap, it is useful to divide markets into three types. Type 1 markets have a mix of both Big Six firms and the other firms. Type 2 markets have only the other firms, and Type 3 markets have only Big Six firms. Figure 9 plots the share of markets with binding maximum price caps across the three types of markets and the three time periods.

In the Learning Period of 2009-2012 depicted by the first bar cluster of Figure 9, Type

1 markets with a mix of firms were the most common (417 markets, 73% of all markets) and had the lowest share with binding price caps, at 10% (first bar cluster, blue bar). By contrast, Type 3 markets with only Big Six firms were much less common (93 markets, 16% of all markets), and were much more likely to reach the maximum price caps, with 53% of markets having a binding maximum price cap (first bar cluster, green bar). Type 2 with only the other firms included 65 markets (11% of all markets), of which 30% had binding price caps (first bar cluster, orange bar).

In the Stable Period of 2013-2016, the share of markets with binding maximum price caps increased across all market types: From 10% to 27% in Type 1 (second bar cluster, blue bar), from 30% to 41% in Type 2 (second bar cluster, green bar), and from 53% to 68% in Type 3 markets (second bar cluster, orange bar).

After the narrowing of the price band in 2017, the share of markets having binding price caps declined across all market types. The share went from 27% to 13% in Type 1 (third bar cluster, blue bar), from 41% to 8% in Type 2 (third bar cluster, green bar), from 68% to 51% in Type 3 (third bar cluster, orange bar).

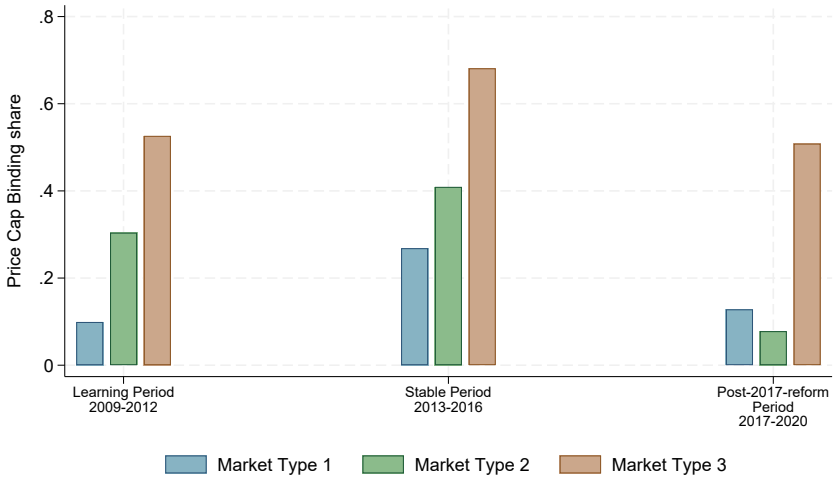
Figure 9 shows that binding maximum price caps were particularly prevalent in markets with only Big Six firms (Type 3), and their prevalence declined after the 2017 narrowing of the price band. For all market types, the share of markets with binding maximum price caps increased from the Learning Period of 2009-2012 to the Stable Period of 2013-2016 and then declined in the Post-2017 Period.

5.3 Reference Price and the Number of Firms

Because a price-band-sized reference-price increase requires all firms in the market to use focal-point pricing, the number of firms in a market may affect the probability of a price-band-sized reference-price increase. We next study how the number of firms in a market is associated with market-level outcomes related to focal-point pricing, including reference-price increases and markets reaching the maximum price cap. In Table 2, we present the probability of price-band-sized reference-price increases and binding maximum price caps by the number of firms (Panel A) and the number of firms outside of the Big Six (Panel B). We exclude the Learning Period and focus on the Stable Period of 2013-2016 and the Post-2017-reform Period of 2017-2020.

In Panel A, the first row shows the shares of markets in the Stable Period of 2013-2016 having reference-price increases equal to the price band. These shares are conditional on the maximum price cap not binding as otherwise the reference price cannot increase. During the Stable Period of 2013-2016, in markets with 3 or more firms, firms succeeded in having

Figure 9: Share of Markets with Binding Maximum Price Caps



Notes: Unit of observation: Reference-price-group-by-quarter. Markets with monopoly products are excluded. Type 1 markets have a mix of both Big Six firms and the other firms. Type 2 markets have only the other firms. Type 3 markets have only Big Six firms. Data source: Fimea (2009–2020).

price-band-sized reference-price increases at least 22% of the time (First row, Column "3", "4", etc.). For all markets, this share is over 26% (First row, Column "All"). The second row shows the same shares in the Post-2017-reform Period. The overall share (Column "All") declined from 26% in the Stable Period to 14% in the Post-2017-reform Period. Conditional on the number of firms, the shares were larger in 2013-2016 than in 2017-2020 irrespective of the number of firms (Panel A, first and second rows).

In Panel A, the third and fourth rows show the shares of markets having binding maximum price caps. The overall share ("All" Column) declined from 51% in the Stable Period to 34% in the Post-2017-reform Period. Conditional on the number of firms, the shares were larger in 2013-2016 than in 2017-2020 irrespective of the number of firms (Panel A, third and fourth rows). In both rows, the share of markets having binding maximum price caps was decreasing in the number of firms.

In Panel B, the first and second rows show the shares of markets having price-band-sized reference-price increases. These shares are conditional on the maximum price cap not binding. The third and fourth rows show the shares of markets having binding maximum price caps. In all rows, the shares of markets having price-band-sized reference-price increases or having binding maximum price caps were decreasing in the number of firms outside the Big Six. These shares were the largest when the market contained only Big Six firms (the "0" Column). The difference in these shares is particularly pronounced between one and two

firms outside the Big Six. Especially in 2017-2020, there is a notable decrease in the share of reference price groups with binding price caps from 56% to 20%.

Table 2: Share of Markets with Price-Band-Sized Reference-Price Increases or Binding Maximum Price Caps

Panel A		**Number of Firms**						
	Years	All	1	2	3	4	5	5+
PB-Sized RP Increase	2013–2016	.26	.28	.19	.22	.36	.32	.23
	2017–2020	.14	.18	.18	.16	.15	.11	.05
Binding Maximum Price Cap	2013–2016	.51	.82	.71	.5	.4	.27	.07
	2017–2020	.34	.71	.48	.27	.18	.06	.01

Panel B		**Number of Firms outside the Big Six**							
	Years	All	0	1	2	3	4	5	5+
PB-Sized RP Increase	2013–2016	.26	.39	.34	.25	.23	.19	.12	.06
	2017–2020	.14	.29	.23	.13	.07	.06	.02	.02
Binding Maximum Price Cap	2013–2016	.51	.85	.7	.41	.15	.03	0	0
	2017–2020	.34	.63	.56	.2	.06	.04	0	0

Notes: Unit of observation: Reference-price-group-by-quarter. PB-Sized RP Increase: Price-band-sized reference-price increase. The PB-Sized RP Increase rows are conditional on markets without binding maximum price caps. Data Source: AFP (2013–2020), Hila (2013–2020).

5.4 Descriptive Regressions of Reference Price Changes

As the reference price depends on the minimum price in the market, a reference-price increase requires all firms in the market to have prices higher than those at the beginning of the previous quarter. We showed that the Big Six frequently used focal-point pricing and increased their prices from quarter to quarter. Therefore, Type-3 markets (only Big Six) were more likely to have a reference-price increase than other markets. In this subsection, we run the following descriptive regressions to check if the probability of a reference-price increase statistically differs between market types:

$$RPIncrease_{mt} = \alpha_0 + \alpha_1 Type2_{mt} + \alpha_2 Type3_{mt} + \beta_0 RPRatio1_{mt} + \beta_1 RPRatio2_{mt} + \beta_2 RPRatio3_{mt} + \gamma Big6Ratio_{mt} + \delta_0 Market_m + \delta_1 Time_t \quad (1)$$

The dependent variable $RPIncrease_{mt}$ is a dummy (0/1) indicating whether market m has a reference-price increase in quarter/year t. α_0 is the constant.

To show which market type is associated with the probability of a reference-price increase,

the first group of explanatory variables consists of last quarter's market type dummies (0/1) $Type2_{mt}$ and $Type3_{mt}$ indicating whether the market consisted of only the other firms (Type 2 markets) or only Big Six firms (Type 3 markets). These variables are mutually exclusive. Type 1 markets (mix of both firms) are omitted as the baseline. The coefficients of these variables thus capture the differences in the probability of a reference-price increase between Type 1 and Type 2, and between Type 1 and Type 3 markets, holding other things constant.

To account for the relation of the previous reference price and following reference-price increases, the second group of explanatory variables $RPRatio1_{mt}$, $RPRatio2_{mt}$, $RPRatio3_{mt}$ are continuous variables equal to the last quarter's reference price divided by the maximum price cap conditional on last quarter's market type. These variables are mutually exclusive and have the max value of 1. Holding other things constant, negative coefficients of these variables would indicate that the market is less likely to have a reference-price increase when the last quarter's reference price is closer to the maximum price cap.

To study how the market structure is correlated with reference-price increases, we include the last explanatory variable $Big6Ratio_{mt}$, which is the share of products from the Big Six in the market. This ratio is only relevant for Type 1 markets where there are both Big Six and the other firms in the market. In Type 2 and Type 3 markets, there is no variation in the share of Big Six's products. Holding other things constant, a positive coefficient of this variable would indicate that a Type 1 market is more likely to have a reference-price increase when the market has a higher share of Big Six products in the last quarter.

Table 3 presents the results of descriptive regressions. We report robust standard errors in parentheses. We have three specifications: The first one only has market type dummies (Panel 1), the second one adds the reference price ratios (Panel 2), and the third one adds the Big Six ratio (Panel 3).

In the first specification, Panel 1 of the market types shows that Type 3 markets (only Big Six) are 29% (Panel 1, Row 2, Column 1) more likely to have a reference-price increase than the baseline Type 1 markets (mix of firms). Type 2 markets are 2% (Panel 1, Row 1, Column 1) more likely to have a reference-price increase than Type 1 Markets. In the second specification, after controlling for the price ratios, the difference between Type 3 markets and Type 1 markets is 57% (Panel 1, Row 2, Column 2), and the difference between Type 2 markets and Type 1 markets is 12% (Panel 1, Row 1, Column 2). These numbers in the third specification after controlling for the price ratios and the Big Six ratio are 83% (Panel 1, Row 2, Column 3) and 27% (Panel 1, Row 1, Column 3) respectively.

In the second and third specification, we control for the price ratios and report the associated coefficients in Panel 2. We find that across market types, being close to the maximum price cap (a higher reference price/price cap ratio) is associated with a lower

probability of a reference-price increase. The estimated coefficients of the reference price ratios for Type 1, Type 2, Type 3 markets are -.38, -.50, -.74 respectively in both second and third specification (Panel 2, column 2 and column 3). These results are intuitive because there is less room for further increases when the reference price is already close to the maximum price cap. The average price cap ratios for Type 1, Type 2, and Type 3 markets are .63, .81, and .74 respectively. From the second column, the average probability of a reference-price increase in Type 1 markets is 25% (= .49 (constant) -.38 (Type 1 coefficient) × .63 (Type 1 average ratio)). This number for Type 2 markets is 21% (= .49 (constant) + .12 (Type 2 dummy) -.50 (Type 2 coefficient) × .81 (Type 2 average ratio)). The average probability of a reference-price increase in Type 3 markets is much higher at 51% (= .49 (constant) +.57 (Type 3 dummy) -.74 (Type 3 coefficient) × .74 (Type 3 average ratio)).

In the third specification, we also control for the Big Six ratio and report the associated coefficient in Panel 3. The positive estimated coefficient of .41 shows that a larger Big Six presence is associated with a higher probability of a reference-price increase. This result suggests that in markets with both types of firms, greater exposure to Big Six firms is related to more increases in the reference price. The estimated coefficients of the price ratios in Panel 2 are not sensitive to the inclusion of the Big Six ratio in the third specification.

Taken together, our descriptive regression results show that the reference price is statistically more likely to increase when a market contains only Big Six firms, when the reference price is clearly below the maximum price cap, and when a mixed market has a large Big Six presence.

6 Structural Model

Building on the findings of the previous descriptive analyses, we estimate a structural model to study firms' pricing incentives and whether observed prices can be explained by competitive pricing or potential coordination. We first estimate a discrete-choice demand model. Then, on the supply side, we distinguish between two periods: The first is the reference-price-setting period at the beginning of each quarter, when firms' prices determine the new reference price. The second consists of the fixed-reference-price periods during the remainder of the quarter, after the reference price has already been fixed. In the reference-price-setting periods, we check for potential coordination by examining whether a firm had incentives to submit a sufficiently low price to become the only product eligible for full reimbursement and pharmacy recommendation. In the fixed-reference-price periods, we check if the fixed reference price prevented firms from pricing higher.

Table 3: Descriptive Regressions of Reference Price Increases

	(1) RP Increase (0/1)	(2) RP Increase (0/1)	(3) RP Increase (0/1)
Constant	.23	.49	.28
Panel 1: Market Types			
Type 2 (Only the other firms)	.02	.12***	.27***
	(.02)	(.04)	(.04)
Type 3 (Only Big Six)	.29***	.57***	.83***
	(.02)	(.05)	(.06)
Panel 2: Reference Price / Price Cap Ratios			
Reference Price Ratio - Market Type 1 (Mix of firms)		-.38***	-.38***
		(.02)	(.02)
Reference Price Ratio - Type 2 (Only the other firms)		-.50***	-.50***
		(.05)	(.05)
Reference Price Ratio - Type 3 (Only Big Six)		-.74***	-.74***
		(.07)	(.07)
Panel 3: Big Six Ratio			
Big Six Ratio - Type 1 (Mix of firms)			.41***
			(.04)
Quarter/Year FEs	Yes	Yes	Yes
Market FEs	Yes	Yes	Yes
F-test of Added Vars (compared to (1))		181.56***	166.61***
Observations	17654	17654	17654

Notes: Unit of observation: Reference-group-by-quarter. We exclude markets with binding price caps and monopoly markets. Robust SEs in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Data source: Fimea (2009–2020).

6.1 Demand

We estimate a discrete-choice demand model that captures substitution across similar products within the same market. Our market definition follows the regulator's definition of a reference price group. It includes all products with the same active ingredient–dosage form–strength–package-size group and an outside good. We assume that there is no substitution across markets. We model product demand as the outcome of a joint decision by physicians, pharmacists, and patients. The indirect utility for consumer i from buying product j in market m (omitting the time period t) is given by:

$$v_{ij} = \alpha + \gamma R_j - \beta_1 ln(P_j|R_j = 1) - \beta_2 ln(P_j|R_j = 0) + \kappa X_j + \mu_m + \xi_j + \epsilon_{ij} \tag{2}$$

where α is the constant, R_j is an endogenous dummy of whether product j is recommended by pharmacists because it is priced at or below the reference price and is reported to be available for all pharmacies, and γ thus measures the increase in demand from being priced at or below the reference price, keeping prices fixed.

P_j is the price of product j, β_1 is the coefficient of log price conditional on the product being priced at or below the reference price, β_2 is the coefficient of log price conditional on the product being priced above the reference price. We have two different price coefficients to account for the difference in consumers' reimbursement between the two groups of products.

X_j includes product characteristics such as being a branded product or a parallel import, and κ is the associated vector of two coefficients. Being a generic product is omitted as the baseline, μ_m is the market fixed effect and ξ_j is the product unobserved quality, which is observed by the consumer but unobserved by the econometrician.

The first three terms $R_j, ln(P_j|R_j = 1), ln(P_j|R_j = 0)$ are endogenous because firms may charge higher prices for products with higher unobserved quality ξ_j.

The consumer and product–specific taste shock ϵ_{ij} is i.i.d across products and follows the EV type 1 distribution. We normalize the utility from the outside good to zero.

The aggregate market share of product j is given by:

$$s_j = \frac{exp(\alpha + \gamma R_j - \beta_1 ln(P_j|R_j = 1) - \beta_2 ln(P_j|R_j = 0) + \kappa X_j + \mu_m + \xi_j)}{1 + \Sigma_j exp(\alpha + \gamma R_j - \beta_1 ln(P_j|R_j = 1) - \beta_2 ln(P_j|R_j = 0) + \kappa X_j + \mu_m + \xi_j)} \tag{3}$$

Because R_j is recorded only for the first month of the quarter in the data, we estimate demand using only data from those months. We are unable to recover R_j for other months based only on observed prices because the pharmacy recommendation status depends on prices and product availability, which we do not observe for the other months. We assume that the demand parameters are the same in the first month and in the remaining months

of the quarter. The key distinction between these periods is that, in the first month, the reference price is determined by the minimum price, whereas in subsequent months, the reference price is fixed. This distinction affects firms' pricing decisions, but it should not directly affect physicians' and consumers' choices among pharmaceutical products.

We estimate demand with price and quantity at the package level for all products in the Finnish pharmaceutical market's reference price system. To address the endogeneity of prices, we construct a Hausman-style price instrument by matching the prices of products in Finland to the prices of the same products in Norway. In the case of not finding a match, we impute prices based on a projection of observed prices in Norway on product characteristics. The intuition of our instrument is that the same product should have the same cost shock across the two countries, but that the demand shocks, conditional on our observables, are independent. Our instruments include the number of products in the market, whether we find a match in Norway, and its interaction with the price instrument.

We estimate the following demand equation with monthly data, assuming an outside good's market share of 5%.[8]

$$ln(s_{jt}) - ln(s_{0t}) = \alpha + \gamma R_{jt} - \beta_1 ln(P_{jt}|R_{jt} = 1) - \beta_2 ln(P_{jt}|R_{jt} = 0) + \kappa X_{jt} + \mu_m + \tau_t + \xi_{jt} \quad (4)$$

where τ_t is the time fixed effect.

We present the results from the logit demand specification (Column 1) and the random coefficient logit demand specifications (Columns 2 and 3) in Table 4. We incorporate a random coefficient with a standard deviation denoted by σ for the dummy of being priced at or below the reference price in Column (2) and for the log price variable for those products priced at or below the reference price in Column (3).[9] For simplicity, we currently use the estimated demand parameters of the logit demand specification in Column (1) for calculations in the following subsections.[10]

All coefficients are precisely measured and of the expected sign. Being priced at or below the reference price and being a branded product lead to higher market shares, while being a parallel–imported product results in lower market shares. Demand is more price elastic for products priced at or below the reference price than for products priced above it.

[8]The share of unfilled prescriptions varies across active pharmaceutical ingredients, the average is 13.3% (Kari et al., 2024). We will conduct robustness checks of varying market share of the outside good.

[9]We only have 1 random coefficient because we are restricted by the number of instruments. We are working on increasing the number of instruments to incorporate more random coefficients.

[10]We plan to use the estimates from the random coefficient logit demand specification for calculations in the following subsections in the near future.

Table 4: Demand Estimation

	(1) Logit	(2) RC Logit	(3) RC Logit
At or below reference price (0/1)	7.83***	9.53***	9.87***
	(.65)	(1.17)	(1.32)
At or below reference price σ		1.93***	
		(.66)	
log(Price) | At or below reference price	-2.01***	-2.14***	-2.31***
	(.23)	(.26)	(.30)
log(Price) σ | At or below reference price			0.65***
			(.24)
log(Price) | Above reference price	-1.22***	-1.30***	-1.33***
	(.11)	(.13)	(.14)
Branded	4.02***	4.67***	4.71***
	(.18)	(.42)	(.44)
Parallel Import	-.81***	-1.57***	-1.67***
	(.09)	(.45)	(.52)
Month FEs	Yes	Yes	Yes
Market FEs	Yes	Yes	Yes
Observations	80171	80171	80171

Notes: Observations are at the product level. The base category is Generic. Robust SEs in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Data source: Fimea (2009–2020).

6.2 Supply

On the supply side, to construct marginal costs, we use each product's minimum observed price over the sample period, adjusted over time using a monthly producer price index for basic pharmaceutical products and pharmaceutical preparations from Statistics Finland. We do not recover marginal costs from firms' first-order conditions for two reasons. The first reason is the presence of corner solutions: Products priced at the reference price have substantially higher sales than products priced just one cent above it. Also, many products are priced at the regulated maximum price caps. The second reason is that we want to explain observed prices with either competitive pricing or potential coordination. As we are unwilling to impose an assumption on firms' pricing behavior, we impose marginal cost assumptions to infer the conduct underlying observed prices. Examples in the recent literature using observed prices as marginal costs include Tunçel (2025) and Dubois et al. (2022). Tunçel (2025) creates bounds for marginal costs based on observed minimum prices and zero marginal cost assumption. Dubois et al. (2022) use the prices of generics as the marginal costs in the US pharmaceutical market.

To model firms' pricing decisions, we distinguish between the beginning of the quarter, when the reference price is determined, and the remainder of the quarter, after the reference price is fixed. At the beginning of the quarter, firms' prices affect the reference price for the whole quarter. Therefore, their profit maximization problem at the beginning of the quarter also includes the profits in the second and third months of the quarter.

$$\max_{P_{j,t}}(P_{j,t} - c_{j,t}) \times s_{j,t} + \sum_{t'=2}^{3}(P_{j,t'} - c_{j,t'}) \times s_{j,t'} \tag{5}$$

When the reference price is fixed in the second and third month of the quarter, the profit maximization problem in month t is the following:

$$\max_{P_{jt}}(P_{jt} - c_{jt}) \times s_{jt} \tag{6}$$

6.3 Reference-price-setting Periods

In this subsection, we focus on prices at the beginning of the quarter and examine whether a firm had incentives to submit a sufficiently low price to become the only product eligible for full reimbursement and pharmacy recommendation. We then compare these unilateral deviation incentives with observed prices. If such a downward deviation were profitable but the firm did not deviate, we interpret the observed pricing behavior as potential coordination.

We calculate the deviation incentives as follows. Although the reference-price-setting

mechanism is designed as a first-price sealed-bid auction, our descriptive evidence shows substantial persistence in firms' prices: Firms submitted prices that differed from their prices at the end of the previous quarter in only about 13% of observations. Assuming that all products submitted prices equal to their prices at the end of the previous quarter, we calculate the expected market share $\widehat{s_{j,t}}$ in the first month if a firm deviated and chose the price $\widehat{P_{j,t}} = min_{k \neq j}\widehat{P_{k,t}} - PriceBand - \epsilon$ at the beginning of the quarter.

A firm considering such a deviation must also take into account its effect on the reference price for the remainder of the quarter. A sufficiently large downward deviation lowers the reference price and can therefore induce subsequent price adjustments by other products. In each deviation scenario, we assume that the prices of all products during the remainder of the quarter would have declined by the same amount as the reference price, while their market shares in the second and third months of the quarter $s^*_{j,t'}$ would remain unchanged. We abstract from punishment or other strategic responses in subsequent quarters because we find little evidence of punishment (more details in Appendix Section A-3). We interpret our classification of potential coordination as an upper-bound estimate.

A firm has an incentive to unilaterally deviate if:

$$(P^*_{j,t} - c_{j,t}) \times s^*_{j,t} + \sum_{t'=2}^{3}(P^*_{j,t'} - c_{j,t'}) \times s^*_{j,t'} < (\widehat{P_{j,t}} - c_{j,t}) \times \widehat{s_{j,t}} + \sum_{t'=2}^{3}(\widehat{P_{j,t'}} - c_{j,t'}) \times s^*_{j,t'} \quad (7)$$

where the deviation price is

$$\widehat{P_{j,t}} = min_{k \neq j}\widehat{P_{k,t}} - PriceBand - \epsilon \quad (8)$$

On the left-hand side of inequality 7, $P^*_{j,t}$ is the observed price of product j in month t, $c_{j,t}$ is the marginal cost of product j in month t, $s^*_{j,t}$ is the observed market share of product j in month t, $\sum_{t'=2}^{3}(P^*_{j,t'} - c_{j,t'}) \times s^*_{j,t'}$ is the total profit of product j in the second and third month of the quarter calculated from the observed prices and the observed market shares in the second and third month of the quarter.

On the right-hand side of inequality 7, $\widehat{P_{j,t}}$ is the deviation price of product j in month t, $\widehat{s_{j,t}}$ is the expected market share of product j in month t given the deviation price and the expected prices of all other firms $\widehat{P_{k,t}}$ in month t, $\sum_{t'=2}^{3}(\widehat{P_{j,t'}} - c_{j,t'}) \times s^*_{j,t'}$ is the expected total profit of product j in the second and third month of the quarter calculated from the expected prices following the deviation and the observed market shares in the second and third month of the quarter.

Table 5 reports, for products observed at the beginning of each quarter, whether our calculations suggest a profitable deviation and whether the product actually deviated in the

data. The first panel, which pools all products, shows that 13.9% (sum of Columns 1 and 2, Panel 1) of products deviated, and 94% of these deviations were profitable (13.0% in Column 2 divided by the sum of 13.9%). The remaining 86.1% of products (sum of Columns 3 and 4, Panel 1) did not deviate, even though for 60% of them (51.5% in Column 4 divided by the sum of 86.1%), a deviation were profitable.

Table 5: Observed Deviations and Profitability of Deviations

	Deviation		No Deviation	
	Not Profitable (1)	Profitable (2)	Not Profitable (3)	Profitable (4)
Panel 1: Overall	**0.9%**	**13.0%**	**34.6%**	**51.5%**
Panel 2: Firm Types				
Big 6	0.3%	5.1%	36.4%	**58.2%**
Other firms	1.4%	20.4%	32.9%	**45.3%**
Panel 3: Market Types				
Type 1 – Big 6	0.3%	5.1%	37.2%	**57.4%**
Type 1 – Non-Big 6	1.4%	18.6%	33.7%	**46.3%**
Type 2 – Non-Big 6	1.6%	28.2%	29.1%	**41.1%**
Type 3 – Big 6	0.5%	5.5%	25.2%	**68.9%**
Panel 4: Time Period				
Learning Period (2009-2012)	1.4%	**9.3%**	53.0%	**36.3%**
Stable Period (2013-2016)	0.8%	**10.9%**	28.1%	**60.3%**
Post-2017-reform Period (2017-2020)	0.6%	**16.7%**	27.4%	**55.3%**

Notes: Deviation: Observed price is more than the price band lower than rivals' prices at the end of the previous quarter. Profitable: Pricing at more than the the price band lower than rivals' prices at the end of the previous quarter is more profitable than keeping the price at the end of the previous quarter. Data source: Fimea (2009–2020).

In the second panel where the rows condition on firm types, the share of Big Six products classified as potential coordination (because they had a profitable deviation but did not deviate) is 58.2% (Panel 2, Row 1, Column 4) while for the other firms, the share is 45.3% (Panel 2, Row 2, Column 4). This result is consistent with the descriptive analysis that the Big Six were more likely to use focal-point pricing and less likely to undercut the previous reference price.

The third panel reports the results by market type. In Type 1 markets with both types of firms, Big Six firms have a higher share of suspected coordination at 57.4% (Panel 3, Row 1, Column 4) compared to 46.3% (Panel 3, Row 2, Column 4) for the other firms. The suspected coordination share is highest in Type 3 markets (only Big Six) at 68.9% (Panel 3, Row 4,

Column 4) while this share is only 41.1% (Panel 3, Row 3, Column 4) in Type 2 markets (only other firms). These patterns are consistent with the descriptive regression results that firms are more likely to succeed in having price-band-sized reference-price increases when the market contains only the Big Six firms, or when the market has a large presence of the Big Six firms.

We next analyze the fourth panel to compare the three time periods. The share of products with a profitable deviation in the Learning Period of 2009-2012 was 45.6% (sum of Columns 2 and 4, Panel 4, Row 1). 20% of them actually deviated (9.3% in Column 2 divided by the sum of 45.6%).

In the Stable Period of 2013-2016, the share of products with a profitable deviation was 71.2% (sum of Columns 2 and 4, Panel 4, Row 2) compared to 45.6% in the previous period. Only 15% of them (compared to 20% in the previous period) actually deviated (10.9% in Column 2 divided by the sum of 71.2%). These results reflect two observations we made in the previous sections. First, firms had stronger incentives to deviate in the Stable Period because the price level had increased substantially compared to the Learning Period. Second, firms deviated less often in relative terms during the Stable Period of 2013-2016 because they frequently used focal-point pricing.

In the Post-2017-reform Period, the share of products with a profitable deviation was 72% (sum of Columns 2 and 4, Panel 4, Row 3). 23% of them (compared to 15% in the previous period) actually deviated (16.7% in Column 2 divided by the sum of 72%). This result indicates that firms deviated more often when the narrower price band was in place after 2017. Our calculations provide an intuitive explanation for this result: It was easier to undercut rivals sufficiently to push them above the reference price (and get the exclusive reimbursement and pharmacy recommendation status) when the price band was €0.5 than when it was €1.5 or €2.

6.4 Fixed-reference-price Periods

In this subsection, we focus on the fixed-reference-price periods during the remainder of the quarter (the second and third months of each quarter) to examine whether the fixed reference price prevented firms from pricing higher. Such a result would provide evidence that the reference price system was effective in restricting prices. To achieve that goal, we ask the following questions: What if the full reimbursement and pharmacy recommendation status had applied to all products instead of only those priced at or below the reference price? How would products currently priced at the reference price have changed their prices? We conduct two analyses. First, we ask whether these products would have unilaterally raised

their prices above the reference price. Second, we ask whether they would have unilaterally raised their prices to the maximum price cap.

For the first analysis, we evaluate the derivative of a product's profit with respect to its own price, holding all rivals' prices fixed. If this derivative is positive for a product priced at the reference price, pricing at the reference price is a corner solution induced by the reference price regulation. In that case, the product would have unilaterally increased its price above the reference price if it had retained its reimbursement and pharmacy recommendation status and hence the reference price restrained the product's price. The derivative is given by:

$$\frac{\partial \Pi_{jt}}{\partial P_{jt}} = s_{jt} + (P_{jt} - c_{jt}) \times \frac{\partial s_{jt}}{\partial P_{jt}} = s_{jt} + (P_{jt} - c_{jt}) \times \frac{(-\beta_1) s_{jt} \times (1 - s_{jt})}{P_{jt}} \tag{9}$$

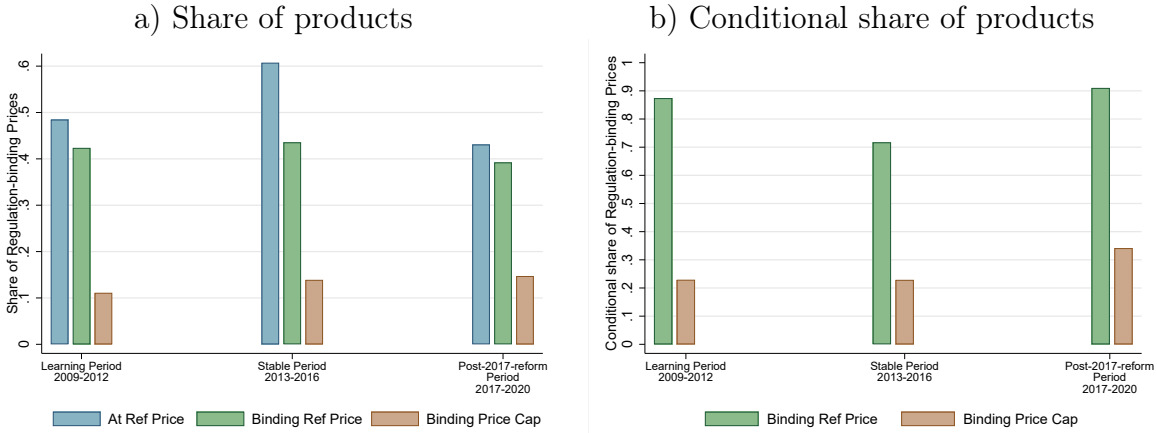
where Π_{jt} is the profit of product j in month t, P_{jt} is the price of product j in month t, s_{jt} is the market share of product j in month t, c_{jt} is the marginal cost of product j in month t.

For the second analysis, we also assume that the rivals' prices are fixed. We then change the price of a product priced at the reference price to the maximum price cap. After calculating the new market share and the new profit, we evaluate the derivative of the product's new profit with respect to its new price. If this derivative is nonnegative, the product would have unilaterally increased its price to the maximum price cap if it had retained its reimbursement and pharmacy recommendation status.

Figure 10 examines whether the clustering of prices at the reference price in the fixed-reference-price periods reflects a binding regulatory constraint. In the left graph, the blue bars show the shares of products priced exactly at the reference price. The green bars show the absolute shares (in the left graph) or the conditional shares (in the right graph) of products having a binding reference price. The orange bars show the absolute shares (in the left graph) or the conditional shares (in the right graph) of products having a binding maximum price cap. The orange bars are lower than the green bars by design. The left graph shows the share of products across these different measures in three time periods. The right graph shows the share of products conditional on being priced exactly at the reference price, also by time period.

In the left graph, in all three time periods, a substantial fraction of products were priced exactly at the reference price, with the share around 40-60 percent (blue bars in the left graph). Also in the left graph, the green and orange bars remain stable across the three time periods. Around 40% (left graph, green bars) of products were priced exactly at the reference price and had a binding reference price. They would have unilaterally charged a higher price than the reference price if they had retained their reimbursement and pharmacy

Figure 10: Fixed-reference-price Prices



Notes: At Ref Price: Priced exactly at the reference price. Binding Ref Price: Priced exactly at the reference price because of corner solution. Binding Price Cap: Priced exactly at the reference price and would be priced at the maximum price cap in the absence of reference prices. Data source: Fimea (2009–2020).

recommendation status. 10-15% (left graph, orange bars) of products priced exactly at the reference price would have unilaterally charged the maximum price cap if they had retained its reimbursement and pharmacy recommendation status. These results provide evidence that during the fixed-reference-price periods, the reference-price system was effective in restraining prices.

The difference between the three time periods is in the conditional shares in the right graph of Figure 10. In the Learning Period of 2009-2012, conditional on being priced at the reference price, 88% of products (right figure, first bar cluster, green bar) would have unilaterally priced above it in the absence of the reference price. This number means that 88% of prices exactly at the reference price can be explained by competitive pricing subject to the reference price constraint. The remaining 12% of products were priced at the reference price even though competitive pricing predicts lower prices. We presume that these firms only chose prices at the reference price within the quarter so that they could signal to others in the market that they would use focal-point pricing at the beginning of the next quarter.

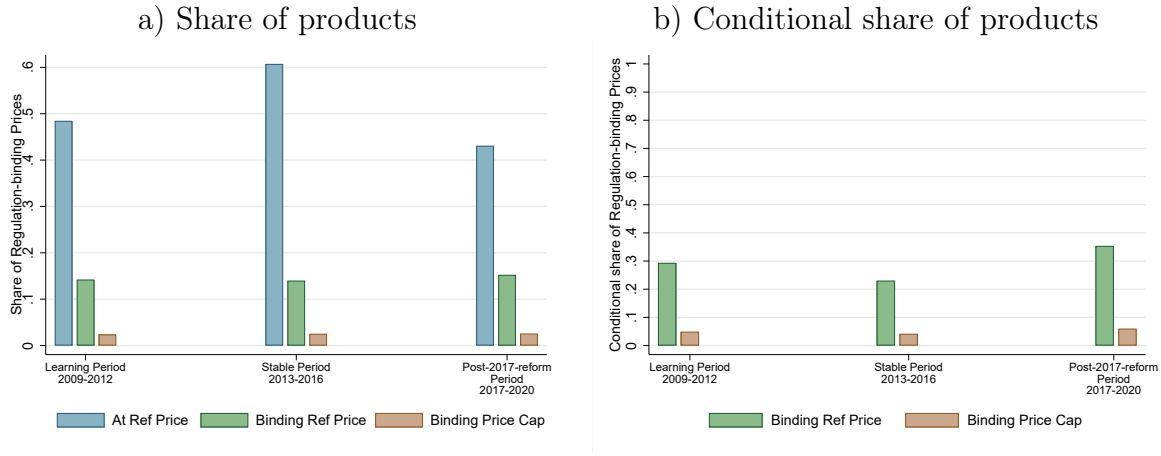
In the Stable Period of 2013-2016, the conditional share of products that were priced exactly at the reference price and had a binding reference price is 71% (right figure, second bar cluster, green bar) compared to 88% in the previous time period. This result indicates that more products (29% compared to 12% previously) potentially tried to signal focal-point pricing during the remainder of the quarter.

In the Post-2017-reform Period, the conditional share of products that were priced exactly at the reference price and had a binding reference price went up to 91% (right figure, third

bar cluster, green bar). A smaller share of firms (9%) potentially trying to signal focal-point pricing is consistent with our descriptive analysis that focal-point pricing became less prevalent after the narrowing of the price band in 2017.

To assess the sensitivity of our conclusions to our marginal cost assumption, we repeat the exercise under the extreme assumption that marginal costs are zero and report the results in Figure 11. Conditional on being priced at the reference price, 23-35% (right graph, green bars) of products would have wanted to price higher if they had retained their reimbursement and pharmacy recommendation status. Conditional on being priced at the reference price, around 5% (right graph, orange bars) of products would have prefered to increase their price to the maximum price cap if they had retained their reimbursement and pharmacy recommendation status.

Figure 11: Fixed-reference-price Prices - Zero MCs



Notes: At Ref Price: Priced exactly at the reference price. Binding Ref Price: Priced exactly at the reference price because of corner solution. Binding Price Cap: Priced exactly at the reference price and would be priced at the maximum price cap in the absence of reference prices. Data source: Fimea (2009–2020).

The two marginal-cost assumptions provide a range for the significance of the reference-price constraint. Even under the extreme assumption of zero marginal costs, the results indicate that the reference price was binding for many products. Our calculations therefore suggest that on the positive side, the reference price constrained firms from increasing prices further in the fixed-reference-price periods.

7 Conclusion

The Finnish pharmaceutical market provides a useful institutional setting for studying how firms learn to adapt to the regulation and how the regulator adjusts the regulation in response. In 2009, the regulator introduced the reference price system that facilitated price competition among firms and led to a substantial drop in prices. However, firms gradually learned to utilize the reference price as a tool for tacit coordination through focal-point pricing. They succeeded in increasing prices during 2012-2016. We find that firms frequently matched the previous quarter's reference price, leading to systematic increases in prices that were equal to the price band. This pattern suggests that price regulation, while restraining prices, may also provide a mechanism for firms to coordinate their prices without explicit communication.

Alarmed by the increase in prices, the regulator learned to adjust the regulation. In 2017, the regulator reduced the price band from €1.5/€2 to €0.5 following a government proposal. The proposal acknowledged that prices had been increasing in some markets, and the main goal in narrowing the price band was to increase price competition (Government of Finland, 2016), but it did not identify focal-point pricing as we document. By the end of our sample in 2022, the narrowing of the price band had resulted in lower prices, and focal-point pricing remained present but less frequent. Future research could examine how firms adapt to the regulation beyond 2022 and whether the decline in focal-point pricing persists over time. Another important question for future research is to determine the optimal size of the price band and the optimal length of the reference price period.

We provide descriptive evidence that a larger price band before 2017 had negative consequences because it determined the size of reference-price increases in many markets with focal-point pricing. We also document a decline in focal-point pricing at the product level and price-band-sized reference-price increases at the market level after the price band narrowing in 2017. Our structural model offers an intuitive explanation for the effectiveness of a narrower price band: It was easier to undercut rivals sufficiently to push them above the reference price (and get the exclusive reimbursement and pharmacy recommendation status) when the price band was €0.5 than when it was €1.5 or €2.

Our results suggest that, although the reference price system facilitated focal-point pricing and reference-price increases at the beginning of the quarter, the fixed reference price prevented many firms from increasing prices higher within the quarter. In addition, the narrower price band introduced in 2017 increased price competition by reducing firms' incentives to use focal-point pricing.

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Appendix

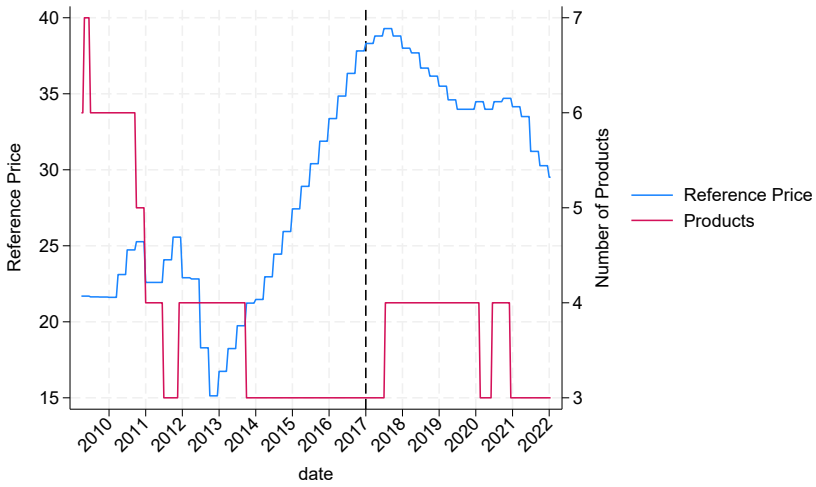
A-1 Additional Graphs

Figure A-1: Average Reference Price



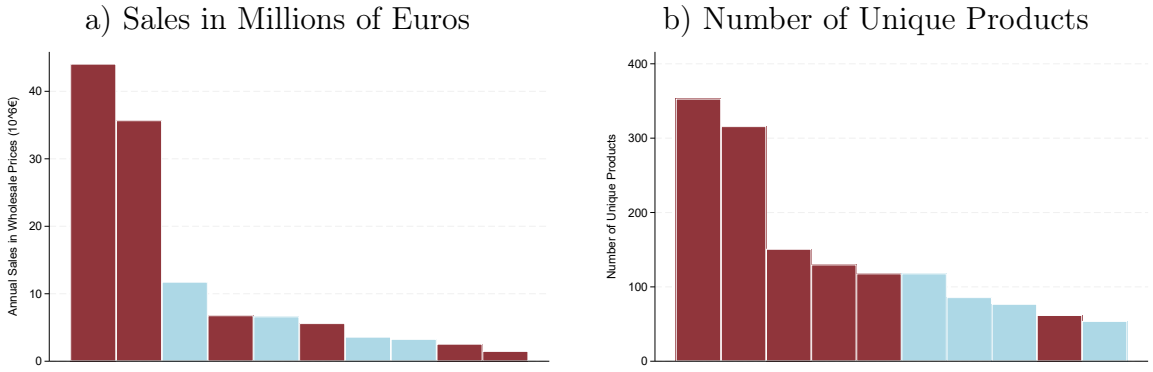
Notes: Weighted by sales in 2016q4. Data source: AFP (2009—2022), Hila (2009–2022), Fimea (2009–2020).

Figure A-2: Prices in a reference price Group



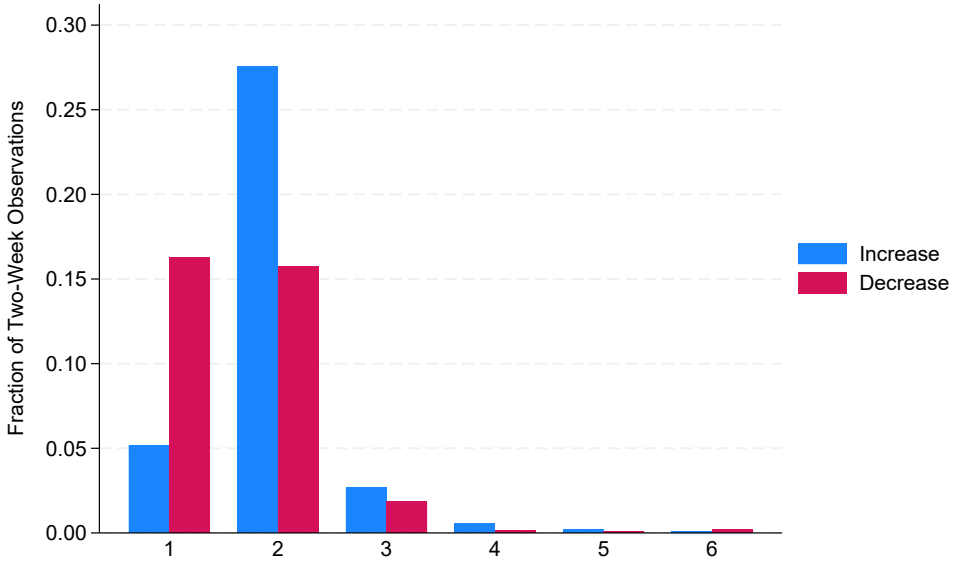
Notes: Data source: AFP (2009–2022).

Figure A-3: Firm-level Annual Sales and Number of Products in 2009 - Top 10



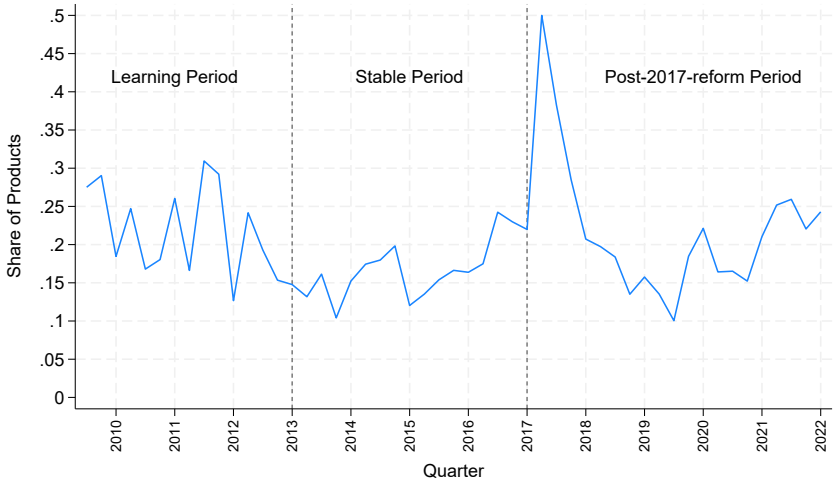
Both figures include the top 10 prominently generic producing pharmaceutical companies in terms of sales or the number of products in the Finnish reference price system. Maroon bars signify the Big Six, blue bars signify other pharmaceutical companies in the top 10. Figure a) represents the annual sales of the Big Six and others in wholesale prices over all products that belong to the reference price system. Figure b) represents the annual number of unique product numbers. Data Source: AFP (2009), Fimea (2009).

Figure A-4: Distribution of the Timing of Price Changes: Other Large Generic-Producing Firms



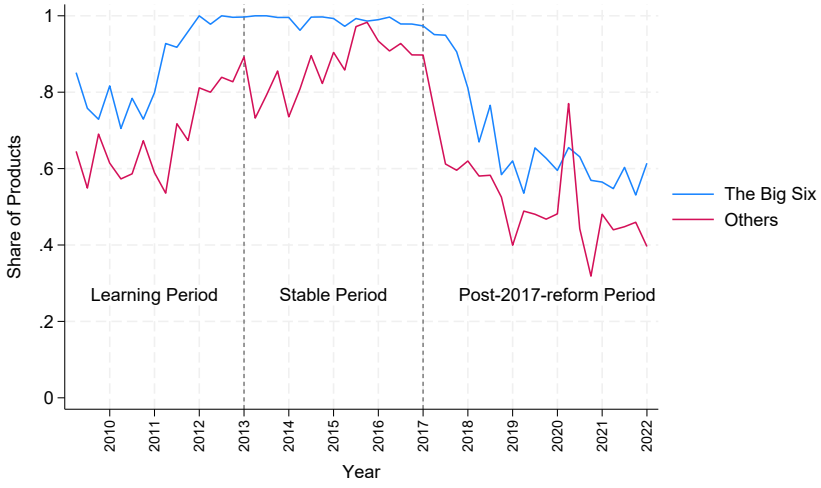
Notes: x-axis: Two-week periods within a quarter. Data source: AFP (2013–2016).

Figure A-5: Share of Products with Reference-Price-Decreasing Prices: Other Large Generic-Producing Firms



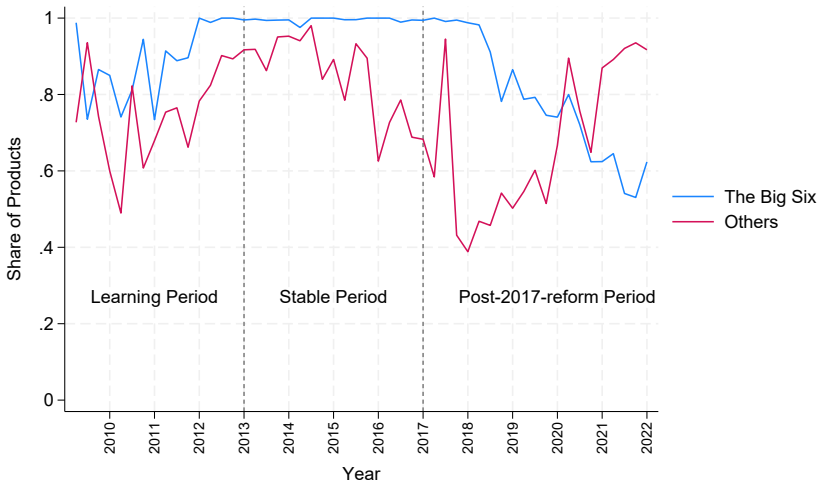
Notes: Share of prices at the beginning of the quarter that would decrease the reference price. Data source: AFP (2009–2022), Hila (2009-2022).

Figure A-6: Share of Products Priced at the Reference Price after a Price Decrease



Notes: Share of products priced at the reference price or at most two cents below it in the second two-week period of a quarter conditional on a price decrease. We exclude reference price groups with (almost) binding price caps. Data source: AFP (2009–2022), Hila (2009-2022).

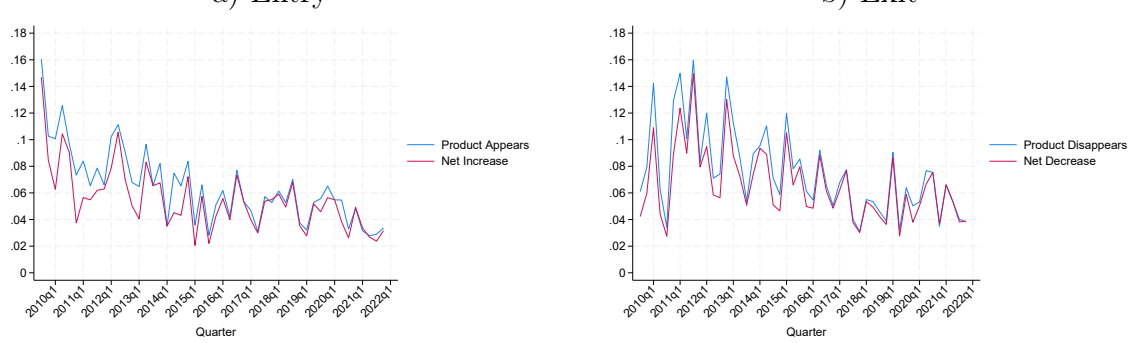
Figure A-7: Share of Products Priced at the Reference Price after a Price Increase



Notes: Share of products priced at the reference price or at most two cents below it in the second two-week period of a quarter conditional on a price increase. We exclude reference price groups with (almost) binding price caps. Data source: AFP (2009–2022), Hila (2009-2022).

A-2 Entry and Exit

Figure A-8: Entry and Exit Patterns



Notes: Share of reference price groups in which an entry or exit occurs. We do not consider formation or removal of entire reference price groups as entry or exit. Data source: AFP (2009–2022), Hila (2009–2022)

Figure A-8 shows entry and exit patterns within the reference price system. The blue line in Figure A-8a represents the quarterly share of reference price groups in which a product number was observed even though it was not observed at the beginning of the previous quarter. Conversely, in Figure A-8b, it represents the quarterly share of reference price groups in which a product number was not observed even though it was observed at the beginning of the previous quarter. The red line describes the share of reference price groups in which there is a net increase in Figure A-8a, or a net decrease in Figure A-8b, in the number of observed product numbers from the previous quarter.

The blue line can be interpreted as an upper bound when we consider entry and exit strictly to and from the reference price system. If a product's product number changes due to a change in e.g., the ownership of the product, the blue line counts it as both an entry and an exit. The red line responds only to changes in the aggregate number of products, acting as an effective lower bound for entry and exit decisions. Despite these differences in the definitions, the lines follow each other closely.

Figure A-8 shows that both entry and exit are high and volatile during the initial learning phase before 2013. Reference price groups experiencing entry range from 4% to 16% per quarter. During the same period, groups facing exits vary between 3% and 16% quarterly. Given the intensity of exits particularly before 2013, the changes in market structure and decreased competition may have further incentivized those firms who stayed in the market to price their products according to focal-point pricing. Afterwards, the share of reference price groups having entrants varies between 2% and 10% each quarter. The corresponding shares for exits are 3% and 12%.

Table A-1 reports OLS regression results for the following specification:

$$\begin{aligned}\mathbb{1}\{RP_{mt} < RP_{mt-1}\} = \beta_0^M + \beta_1^M \times \text{Entry}_{mt} + \beta_2^M \times \mathbb{1}\{Year_{mt} \geq 2017\} \\ +\beta_3^M \times \mathbb{1}\{Year_{mt} < 2013\} + \beta_4^M \times \mathbb{1}\{Entry_{mt}\} \times \mathbb{1}\{Year_{mt} \geq 2017\} \\ +\beta_5^M \times \mathbb{1}\{Entry_{mt}\} \times \mathbb{1}\{Year_{mt} < 2013\} + \lambda_m^M + \lambda_t^M + \epsilon_{mt}^M\end{aligned} \tag{A-1}$$

That is, we regress a dummy indicating a decrease in the reference price on one of our entry measures in Figure A-8 (Entry_{mt}) in reference price group m and quarter t, a dummy indicating Post 2017 time period ($\mathbb{1}\{Year_{mt} \geq 2017\}$), a dummy indicating Pre 2013 time period ($\mathbb{1}\{Year_{mt} < 2013\}$), their interactions, reference price group fixed effects (λ_m^M) in columns 2, 3, 5, and 6, and quarter (λ_t^M) fixed effects in columns 3 and 6. The data is on reference price group and quarter level. We focus on 2009Q3–2020Q4.

In all of the descriptive regressions of Table A-1, entry is positively associated with a decrease in the reference price that quarter regardless of whether we measure it as new product numbers we did not observe at the beginning of last quarter in columns 1–3 or an increase in the aggregate number of products in columns 4–6. The linear probability model predicts an increase of 31 percentage points in the probability of reference price decrease when we do not include reference price group or quarter fixed effects. Once we include these fixed effects and focus on within-group variation, the increase the coefficient implies falls to 16 percentage points. In addition, both before 2013 and after 2017, entry is even more strongly associated with a decrease in the reference price. After 2017, there is a 7–9 percentage point increase in the probability of the reference price decreasing when a new product enters. The result is consistent with hypotheses such as there being a behavioral change or the identity of entrants changing. Similarly, a 7–9 percentage point increase in the probability of the reference price decreasing before 2013 highlights the initial learning phase and potential differing composition of entrants.

Next, we shift attention to individual products and companies. Table A-2 reports OLS estimates using product-quarter level data for 2009Q3–2020Q4. We estimate the following specification:

$$\begin{aligned}\mathbb{1}\{P_{jmt} < RP_{mt-1} - PriceBand_{mt}\} = \beta_0^J + \beta_1^J \times \text{Entrant}_{jmt} \\ +\beta_2^J \times \mathbb{1}\{Year_{jmt} \geq 2017\} + \beta_3^J \times \mathbb{1}\{Year_{jmt} < 2013\} \\ +\beta_4^J \times \text{Entrant}_{jmt} \times \mathbb{1}\{Year_{jmt} \geq 2017\} \\ +\beta_5^J \times \text{Entrant}_{jmt} \times \mathbb{1}\{Year_{jmt} < 2013\} + \epsilon_{jmt}^J\end{aligned} \tag{A-2}$$

The dependent variable is an indicator equal to one if product j in reference price group m sets a price in quarter t low enough to reduce the reference price from the previous quarter. Entrant_{jmt}, indicates whether the product is an entrant. We interact this entrant dummy with dummies for the post-2017 period and the pre-2013 period. Column (1) reports the baseline specification, column (2) adds reference price group fixed effects, and column (3) adds product fixed effects.

These descriptive results indicate that entrants are more likely than incumbents to price below the threshold required to reduce the reference price. In columns (1) and (2), this

Table A-1: Entry, Product Expansion, and Reference Price Decreases

	Entry			Net Increase in Products		
	(1) RP ↓	(2) RP ↓	(3) RP ↓	(4) RP ↓	(5) RP ↓	(6) RP ↓
Entry	0.307***	0.159***	0.155***			
	(0.021)	(0.018)	(0.018)			
Post 2017	0.147***	0.103***		0.148***	0.104***	
	(0.010)	(0.011)		(0.010)	(0.011)	
Entry × Post 2017	0.070**	0.085***	0.084***			
	(0.028)	(0.025)	(0.025)			
Pre 2013	0.035***	0.097***		0.039***	0.099***	
	(0.010)	(0.010)		(0.010)	(0.010)	
Entry × Pre 2013	0.088***	0.088***	0.070***			
	(0.026)	(0.025)	(0.024)			
Net Increase in Products				0.364***	0.204***	0.198***
				(0.022)	(0.019)	(0.019)
Net Increase in Products × Post 2017				0.025	0.054**	0.056**
				(0.030)	(0.027)	(0.027)
Net Increase in Products × Pre 2013				0.079***	0.085***	0.068**
				(0.029)	(0.028)	(0.027)
Constant	0.165***	0.176***	0.242***	0.165***	0.176***	0.242***
	(0.008)	(0.006)	(0.001)	(0.008)	(0.006)	(0.001)
Substitution-group FE	No	Yes	Yes	No	Yes	Yes
Quarter FE	No	No	Yes	No	No	Yes
Observations	35,200	35,172	35,172	35,200	35,172	35,172
R-squared	0.064	0.296	0.352	0.066	0.298	0.354

Standard errors in parentheses

Dependent variable is an indicator for a decrease in the reference price. Standard errors are clustered at the reference price group level. Base levels and omitted variables are not reported. Data source: AFP (2009–2020), Hila (2009–2020)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

tendency is particularly prominent after 2017. However, when we include product fixed effects in column (3), the entrant × post-2017 interaction decreases and becomes statistically insignificant. This suggests that the stronger post-2017 association may partly reflect changes in the composition of entrants rather than changes in the behavior of the same products over time.

The entrant × pre-2013 interaction shows the opposite pattern. It is small and statistically insignificant in columns (1) and (2), but becomes positive and statistically significant after adding product fixed effects in column (3). This is consistent with behavioral differences among entrants between the initial learning phase in 2009–2012 and the main focal-pricing period in 2013–2016.

Table A-2: Entry and Reference-Price Undercutting

	(1) Undercut	(2) Undercut	(3) Undercut
Entrant	0.215***	0.198***	0.116***
	(0.015)	(0.014)	(0.013)
Post 2017	0.097***	0.084***	0.057***
	(0.005)	(0.005)	(0.005)
Entrant × Post 2017	0.133***	0.119***	0.008
	(0.023)	(0.021)	(0.021)
Pre 2013	0.026***	0.037***	0.030***
	(0.004)	(0.004)	(0.003)
Entrant × Pre 2013	0.009	0.008	0.047**
	(0.021)	(0.021)	(0.019)
Constant	0.065***	0.067***	0.081***
	(0.003)	(0.002)	(0.002)
Substitution-group FE	No	Yes	No
Product FE	No	No	Yes
Observations	124,442	124,436	124,239
R-squared	0.033	0.085	0.371

Standard errors in parentheses

Dependent variable is an indicator for pricing sufficiently below the previous reference price to reduce the new reference price. Standard errors clustered at the reference price group level. Omitted and base categories are not reported. Data Sources: AFP (2009–2020), Hila (2009–2020)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A-3 Descriptive Regressions on Pricing after a Decrease in the Reference Price

We estimate OLS regressions with product fixed effects using product and quarter level price data for Table A-3. The dependent variables are dummy variables in all columns. In column (1), the outcome is Price Decrease which equals one if the wholesale price of the product decreases at the beginning of the quarter relative to the price at the end of the previous quarter. In column (2), Focal pricing equals one if the product follows the focal pricing rule. In column (3), Focal Pricing or Higher equals one if the product is priced according to focal pricing or higher than that. Otherwise, the dependent variables equal zero.

The independent variables are Lagged RP Decrease, Stable, Post, And Almost Binding Cap t=-2, all of which are dummy variables. Lagged RP Decrease equals one if the reference price decreased in the previous quarter. Stable equals one in the Stable Period, 2013–2016, while Post equals one in the Post-2017-Reform Period. Binding Cap t=-2 equals one if, two quarters earlier, the reference price was sufficiently close to the price cap that price-band-sized reference-price increases could not occur. This variable and its interactions are meant to control for observations that belong to such groups. Nonetheless, we include the interactions of all of the independent variables in Table A-3.

The main coefficients of interest include Lagged RP Decrease because those describe how products are priced after a decrease in the reference price. We exclude observations in 2014Q2 and 2017Q2 in column (2) due to pharmacy markup and price band reforms mechanically decreasing the reference price in the previous quarter. We exclude these quarters and 2014Q1 and 2017Q1 in columns (2) and (3) due to the same reforms causing mechanical zeros in Lagged RP Decrease in these quarters. We also restrict the sample to observations that were not priced in the previous quarter in a way that would induce a decrease in the reference price.

The most relevant coefficients, Lagged RP Decrease, Lagged RP Decrease $\times$ Stable, and Lagged RP Decrease $\times$ Post, indicate that in the Learning Period in 2009-2012, the probability of prices decreasing between the end of the previous quarter and the beginning of a new quarter increases approximately 2.6 percentage points after a decrease in the reference price in the same reference price group at the beginning of a previous quarter. This relationship is weaker in the Stable Period, at approximately 1 percentage point, and rather similar in the Post-2017-Reform Period. While this finding may be consistent with other firms predicting further price decreases or punishing an earlier undercutter with further reference price decreases, these results may be driven by other factors, for example, group-level demand shocks.

However, columns (2) and (3) provide little evidence of wide-spread one time punishments through deviations from focal pricing. In the Learning Period, column (2) does not indicate a relationship between a reference price decrease in the previous quarter and the following focal pricing. In the Stable and Post-2017-Reform periods, such a decrease is associated positively with focal pricing. Column (3) also indicates a positive relationship between a decrease in the reference price and products other than the potential undercutter being priced according to focal pricing. Therefore, this limited examination of pricing behavior provides little evidence of systematic punishments following decreases in the reference price.

Table A-3: Regressions on Pricing after a Decrease in Reference Price

	(1) Price Decrease	(2) Focal Pricing	(3) Focal Pricing or Higher
Lagged RP Decrease	0.0263***	-0.0170	0.0418***
	(0.00416)	(0.0102)	(0.00947)
Stable	-0.00814*	0.00284	0.0793***
	(0.00319)	(0.0107)	(0.00947)
Lagged RP Decrease × Stable	-0.0163**	0.0799***	0.00603
	(0.00541)	(0.0154)	(0.0106)
Post	-0.0234***	-0.114***	-0.0979***
	(0.00368)	(0.0146)	(0.0142)
Lagged RP Decrease × Post	0.00225	0.0455***	-0.00844
	(0.00517)	(0.0130)	(0.0123)
Binding Cap t=-2	-0.00676	-0.459***	-0.518***
	(0.00356)	(0.0176)	(0.0188)
Lagged RP Decrease × Binding Cap t=-2	0.000554	0.426***	0.451***
	(0.00959)	(0.0246)	(0.0231)
Stable × Binding Cap t=-2	0.00592	-0.00218	-0.0812***
	(0.00362)	(0.0126)	(0.0120)
Lagged RP Decrease × Stable × Binding Cap t=-2	-0.0133	-0.0835*	-0.0165
	(0.0113)	(0.0349)	(0.0327)
Post × Binding Cap t=-2	0.0272***	0.141***	0.0928***
	(0.00446)	(0.0184)	(0.0190)
Lagged RP Decrease × Post × Binding Cap t=-2	-0.0213	-0.0583	0.0143
	(0.0124)	(0.0299)	(0.0292)
Constant	0.0485***	0.556***	0.697***
	(0.00255)	(0.0106)	(0.0107)
Product FE	Yes	Yes	Yes
Observations	115089	110123	110123
R^2	0.228	0.577	0.650

Unit of observation: Product-by-quarter. Standard errors are clustered on reference price group level.
Standard errors in parentheses * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Data Source: AFP (2009-2022), Hila (2009-2022)

A-4 Branded Products

Table A-4 reports the average market share in addition to the 25th, 50th, and 75th percentiles of market shares for branded products conditional on a branded product being present in a reference price group. On average, a branded product accounts for roughly half of sales within a reference price group when it is present. However, there is substantial variation in the market shares across products and markets since the 25th percentile captures only approximately 10% while the 75th percentile exceeds 80%.

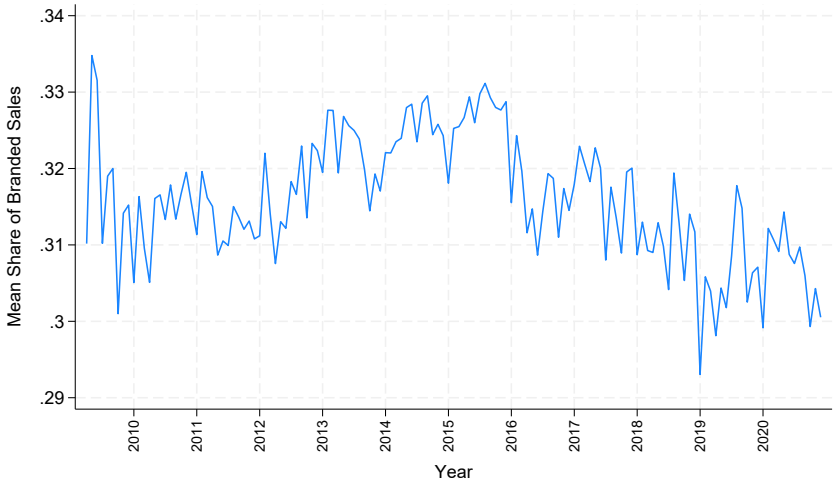
Table A-4: Distribution of Branded Product Market Shares by Period

	Branded product market shares			
Period	Mean	p25	p50	p75
Learning period (2009–2012)	0.460	0.126	0.390	0.815
Stable period (2013–2016)	0.476	0.086	0.436	0.893
Post-2017-reform period (2017–2020)	0.514	0.099	0.571	0.881

Notes: The table reports the mean and the 25th, 50th, and 75th percentiles of branded product market shares across product-month observations. Branded product market share is defined as product-level sales divided by total sales within the substitution group-month, conditional on the product being branded and observed. Data Sources: AFP (2009-2020), Fimea (2009-2020), Hila (2009-2020)

Figure A-9 reports the mean share of branded product sales over all reference price groups. Throughout 2009–2020, the share remains relatively stable between 0.29 and 0.33.

Figure A-9: Mean Share of Branded Product Sales

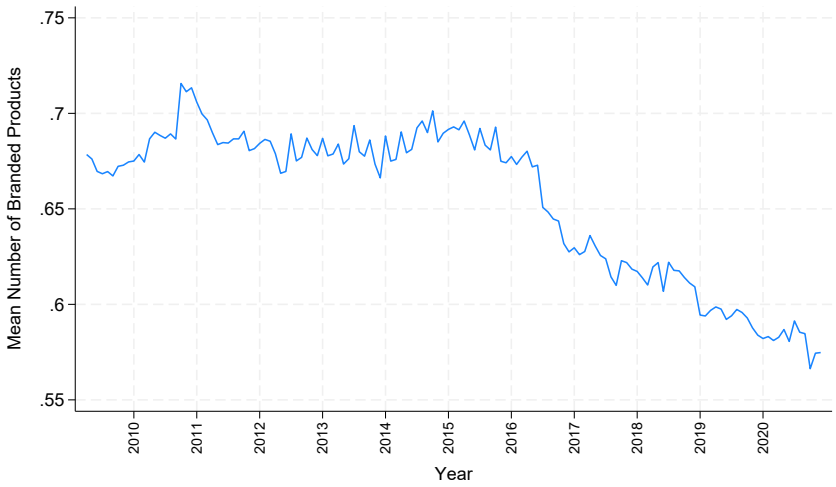


Monthly mean share of branded product sales over reference price groups. Data Sources: AFP (2009-2020), Fimea (2009-2020), Hila (2009-2020)

Figure A-10 reports the mean number of branded products in a reference price groups. On average, there are approximately 0.7 branded products per reference price group before the

price band reform. The share begins to noticeably fall in 2016 towards 0.6 branded products. The beginning of the decreasing trend coincides with a regulatory price cap decrease for branded products in July 2016.

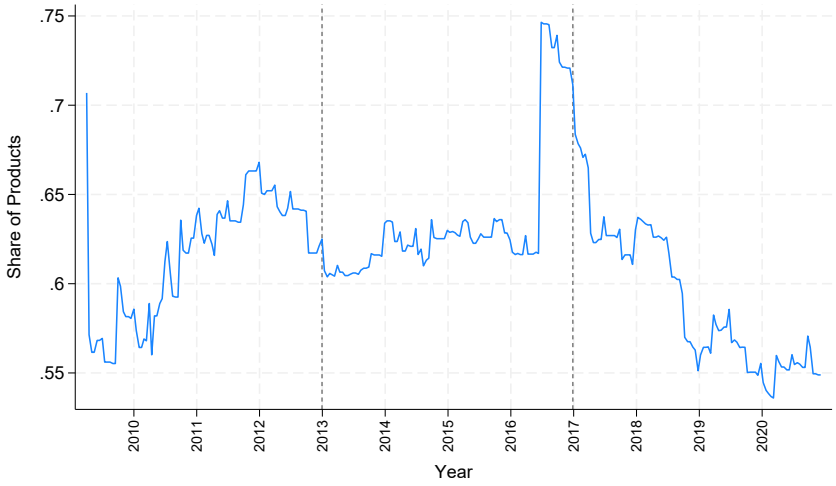
Figure A-10: Mean Number of Branded Products



Monthly mean number of branded products in reference price groups. Data Sources: AFP (2009-2020), Fimea (2009-2020), Hila (2009-2020)

Figure A-11 shows the share of branded products that are priced exactly at their regulatory price cap. The share fluctuates between 55% and 75% over time. The spike in 2016 can be explained by the regulatory decrease in the price caps.

Figure A-11: Share of Branded Products at Price Cap

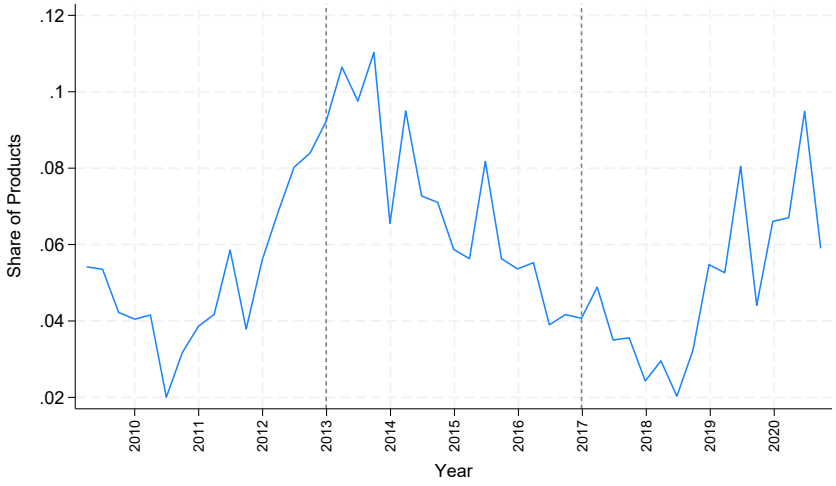


Semimonthly share of branded products at a price cap. Data Sources: AFP (2009-2020), Fimea (2009-2020), Hila (2009-2020)

Figure A-12 reports the share of branded products that are priced exactly at the level that sets the reference, while not being priced at the regulatory price cap. Thus, it is the

share of branded products that are the cheapest or share the exact same price with the cheapest available products at the beginning of a quarter. This share fluctuates between 2% and 11% over time which indicates that branded products play only a limited direct role in determining the reference price.

Figure A-12: Share of Branded Products setting the Reference Price



Quarterly share of branded products at the lower bound of the price corridor (reference price - price band) and not at the price cap at the beginning of a quarter. Data Sources: AFP (2009-2020), Fimea (2009-2020), Hila (2009-2020)

A-5 Event Studies

We estimate event studies at the active-ingredient level using average pharmaceutical expenditure per defined daily dose (DDD) and the number of distinct firm names as the outcomes. Norwegian corresponding active ingredients form the control group. We aggregate outcomes at market-level rather than at the product-level because the price of one product may affect the prices of competing products in the same market. A product-level analysis could violate the Stable Unit Treatment Value Assumption (SUTVA) through interference across products. Following Kortelainen et al. (2023), we define markets on active ingredient level rather than reference price group level. A broader market definition should further alleviate concerns regarding SUTVA although it may dilute the estimated effect due to inclusion of products that are less directly affected by the reform (Hausman, 2001). In addition, because reference price groups are specific for the Finnish institutional setting, mapping across countries is cleaner on a broader, active ingredient, level.

Following Kortelainen et al. (2023) and Markkanen et al. (2026), we construct average expenditure by dividing nominal wholesale sales by the number of defined daily doses sold at the active ingredient level and then taking the logarithm of this variable. We calculate the number of firms based on distinctly named firms within a market.

We construct the control group using the same active ingredients observed in Norway. Because the price band reform applies to all Finnish reference price groups, there are no plau-

sible within-country control groups. We consider Norway a credible control country because pharmaceutical market conditions and expenditure trends appear reasonably comparable to those in Finland during the pre-reform period. Markkanen et al. (2026), for example, document broadly similar effects of loss of exclusivity on average expenditure in Finland and Norway during the early and mid-2010s.

Norway also regulates prices through two maximum price controls. The first price control mechanism is based on the mean price in three countries where the prices of the corresponding products are the lowest among a basket of nine European countries (Weise et al., 2018). Therefore, Norwegian prices, at least when bound by the price cap[11], should reflect changes in European price trends. Norway also uses a stepped price model which imposes predetermined price cuts soon after the product begins to face generic competition (Weise et al., 2018). We do not consider this part of the regulation as a notable issue for identification because it is mainly targeted for products that have recently lost patent protection and most markets in our sample have been under generic competition for quite a while already.

Some markets affected by the 2017 price band reform were also affected by another reform in July 2016. The price caps of some of the branded and parallel imported pharmaceuticals were adjusted to the same level as the price caps of generics in the same reference price group. Whether this reform affected a market was dependent on whether such products were present in the Finnish pharmaceutical market at the time. Because entry and exit decisions are endogenous, selection into exposure to the price cap reform is a concern. Therefore, an event study setting does not allow us to conclusively separate the effects of the price cap reform and the price band reform for those markets that faced both treatments. In addition due to selection, we cannot rule out, and likely should expect, that the price band reform affects differently those markets that faced the price cap reform and those that did not.

Our parallel trends assumption is that without the reforms, the logarithmic market-level outcomes in Finland would have developed with similar trends as their Norwegian counterparts. Because the treatments and their timings are different for those markets that faced the price cap reform and those that did not, the former being affected by the price cap reform after July 2016 and by both price cap and price band reforms after January 2017, we estimate two separate event studies. We set the baseline period to quarter two of 2016 for the markets that were affected by the price cap reform and quarter four of 2016 for those that were not because these were the last quarters before the respective reforms.

We restrict the sample to the period from 2015 through the third quarter of 2018 and aggregate the data at the active-ingredient-by-country-by-quarter level. We use a balanced sample of active ingredients so that we observe sales from each active ingredient in both countries through the whole sample period to avoid compositional changes. However, this restriction rules out, for example, smaller markets due to us restricting the markets only to those that have positive sales each quarter.

We plot raw means of the outcome variables for Finland and Norway separately for the active ingredients that faced the price cap reform and those that did not in Figures A-13 and A-14. The level differences between countries are expected by design in Figure A-13 because we measure average expenditure per DDD in local nominal currency. There are also level

[11]Sales in currency-wise, 56%–63% of quarterly sales are from products that are within NOK 10 (approx €1.1 in our sample period) from their price caps conditional on us observing a price cap for the product.

differences between the two market groups — those markets that faced the price cap reform had higher average expenditure in both countries before the treatment. However, at the end of our sample period, average expenditure converges in Finland between these two groups.

Figure A-13: Logarithmic Average Expenditure Means

a) Price Cap Reform b) No Price Cap Reform

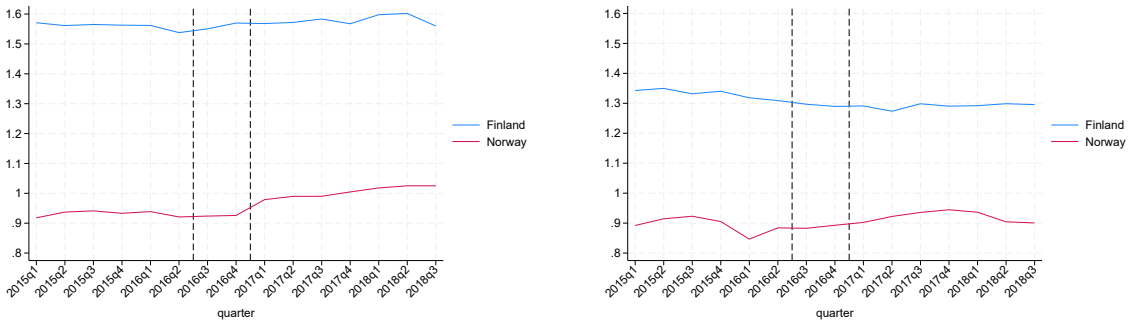


Unweighted mean of logarithmic average expenditure per DDD over markets in Finland and Norway. The balanced sample contains 63 distinct active ingredients observed in both countries in panel a) and 85 in panel b). Data source: AFP (2015–2018), Farmastat (2015–2018), Fimea (2015–2018).

Figure A-14 shows level differences in the number of firms. Finland has more firms per market on average compared to Norway. In addition, markets in Finland that are affected by the price cap reform have more firms than those which do not. Changes in the mean of number of firms are modest in both countries. The number of firms increases slightly in Norway in the post-treatment period, at least relative to 2016 levels.

Figure A-14: Logarithmic Number of Firms Means

a) Price Cap Reform b) No Price Cap Reform



Unweighted mean of logarithmic number of firms over markets in Finland and Norway. The balanced sample contains 63 distinct active ingredients observed in both countries in panel a) and 85 in panel b). Data source: AFP (2015–2018), Farmastat (2015–2018), Fimea (2015–2018).

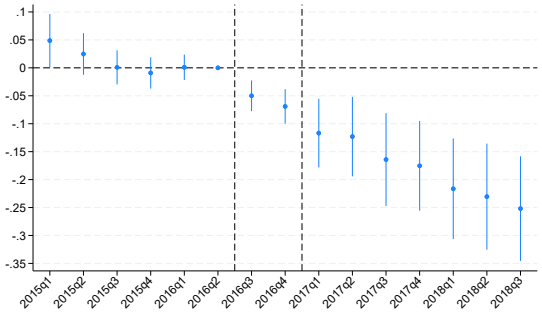
Our estimation equation is as follows:

$$\ln(y_{mct}^{DID}) = \alpha^{DID} + \sum_{k \neq k_0} \beta_k^{DID} \left[\mathbf{1}\{c = \text{Finland}\} \times \mathbf{1}\{t = k\}\right] + \lambda_t^{DID} + \lambda_{mc}^{DID} + \varepsilon_{mct}^{DID}, \quad \text{(A-3)}$$

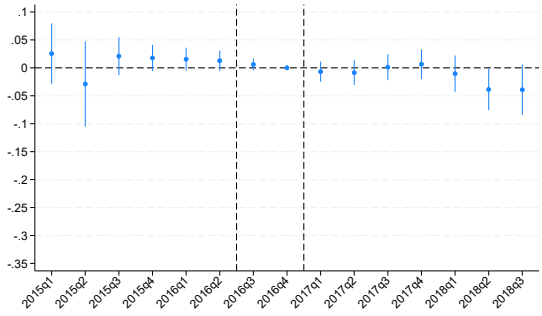
in which y_{tmc}^{DID} denotes the outcome variable in quarter t, market m, and country c. The term α^{DID} is a constant, $\mathbf{1}\{c = \text{Finland}\}$ is an indicator for the treated country, and $\mathbf{1}\{t = k\}$ is an indicator for quarter t. The parameters λ_t^{DID} and λ_{mc}^{DID} denote quarter fixed effects and market-by-country fixed effects, respectively. ϵ_{tmc}^{DID} is the error term. The coefficients β_t^{DID} capture the evolution of the treatment effect over time. We estimate the specification using a two-way fixed-effects estimator. We cluster standard errors on active ingredient level and assign the same weight to all active ingredient.

Figure A-15: Event Study on Logarithmic Average Expenditure

a) Price Cap Reform



b) No Price Cap Reform



The dependent variable is the logarithm of the average expenditure per DDD. The unit of observation is an active-ingredient-by-country-by-quarter. Finland constitutes the treatment group, and Norway the control group. Observations are matched across countries and periods by active ingredient. Standard errors are clustered at the active-ingredient level. The balanced sample contains 63 distinct active ingredients observed in both countries in panel a) and 85 in panel b). Data sources: Fimea (2015–2018), Farmastat (2015–2018), and AFP (2015–2018).

Figure A-15 reports the estimated effects on average expenditure per DDD, β_t^{DID}. In both panels, the pre-treatment coefficients are generally statistically insignificant and close to zero. The estimated average effect on the treated (ATT) of the price cap reform on average expenditure per DDD is -7%[12] by 2016Q4 under the assumption that there is no anticipation due to the price band reform. The following joint effect of the price cap and price band reforms develops gradually from -11% in 2017Q1 to -22% in 2018Q3.

While the effects are not directly separable in this setting, we can form an upper bound for the decrease in average expenditure per DDD due to the price band reform under additional assumptions. If we assume that the effect of the price cap reform does not become less negative and, absent the price cap reform's effect, the rest of the effect can be attributed to the price band reform, the estimated ATT due to the price band reform would be at most -16% by the end of our sample period. Given the trend in the joint effect and our descriptive

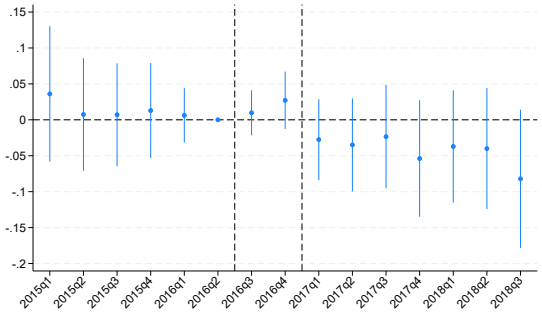
[12] Under the assumption that the error terms are normally distributed and homoskedastic, we transform our estimated coefficients to percentages $(e^{\beta_k^{DID}} - 1) \times 100\%$.

results, such as Figures 1 and A-1, it is also not obvious that the average expenditure per DDD decreasing trend would stop here if we had data from the control group for a longer period.

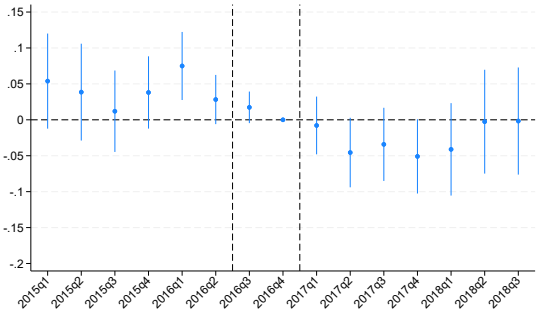
Figure A-15b) shows that the estimated ATT of the price band reform on average expenditure per DDD for the group that did not face the price cap reform is close to zero. The point estimates indicate a decrease of at most -4% by the end of our sample period. This difference between the treatment groups may be indicative of smaller than expected effects of the price band reform on average expenditure per DDD although we find the hypothesis on selection-induced heterogeneous responses a plausible alternative.

Figure A-16: Event Study on Logarithmic Number of Firms

a) Price Cap Reform



b) No Price Cap Reform



The dependent variable is the logarithm of the number of firms. The unit of observation is an active-ingredient-by-country-by-quarter. Finland constitutes the treatment group, and Norway the control group. Observations are matched across countries and periods by active ingredient. Standard errors are clustered at the active-ingredient level. The balanced sample contains 63 distinct active ingredients observed in both countries in panel a) and 85 in panel b). Data sources: Fimea (2015–2018), Farmastat (2015–2018), and AFP (2015–2018).

Figure A-16 reports the estimated effects on the number of firms per market. The coefficients are statistically insignificant and close to zero for the most part both before the treatments and after them. Thus, they suggest that the reduction in average expenditure is not associated with an increase in the aggregate number of firms operating in the market. However, this finding does not rule out the possibility that changes in the composition of the entrants are a plausible mechanism which drives average expenditure and price decreases. If firms pricing their products more aggressively replace incumbents, as potentially suggested by Section A-2, the market-level prices and average expenditure are expected to decrease.

A-6 Back-of-the-envelope calculations - Change of the price band

A-6.1 Back-of-the-envelope calculation 1 - €0.5 price band

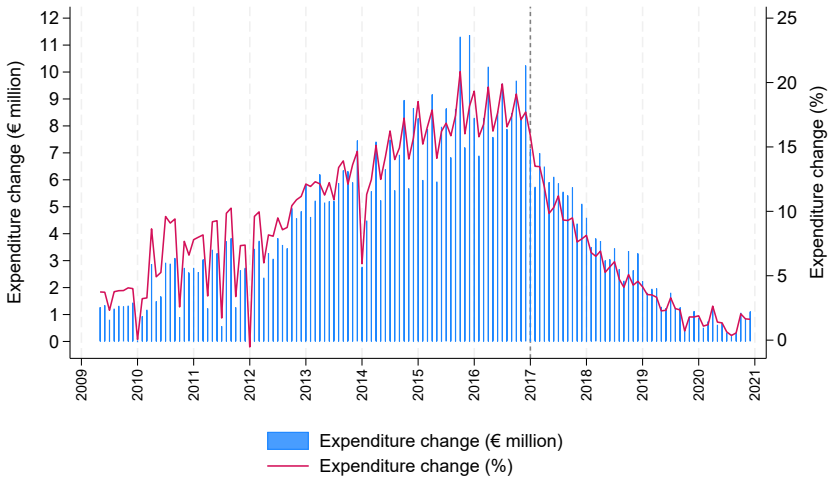
Our first back-of-the-envelope calculation asks what pharmaceutical expenditure would have been if the 2017 reform, which reduced the price band to €0.5, had instead been implemented

in 2009. To construct counterfactual auction prices, we again assume that rivals would submit bids equal to their prices at the end of the previous quarter and calculate a product's incentives to deviate under the counterfactual price band. If deviation is not profitable, we use the product's price from the end of the previous quarter. We do the same when deviation is profitable but the corresponding observation is classified as potentially collusive. If deviation is profitable and the observation is not classified as potentially collusive, we instead assign the product its calculated deviation price.

For non-auction periods, holding other prices fixed, we assume that products whose observed prices lie between the (lower) counterfactual reference price and actual reference price reduce their prices to the counterfactual reference price in order to remain within the price corridor. Using these counterfactual prices, we calculate the resulting change in the total pharmaceutical expenditure.

Figure A-17 shows that introducing the €€0.5 price band in 2009 rather than in 2017 would have reduced monthly pharmaceutical expenditure by as much as €10 million, or approximately 20% of total expenditure. A narrower price band limits the extent to which the reference price can increase from one quarter to the next, even when firms continue to use focal-point pricing. It also reduces the size of the price cut required for a deviating firm to undercut its rivals sufficiently to push them outside the new price corridor, thereby making such deviations more attractive.

Figure A-17: Change in the total expenditure - €0.5 price band



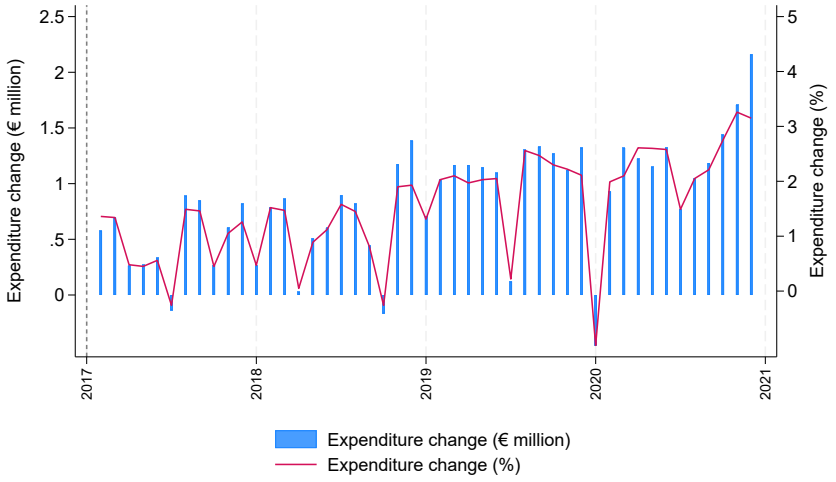
Notes: Data source: Fimea (2009–2020).

The savings accumulate over time, generating a snowball effect as the gap between the simulation and observed price levels widens between 2009 and 2017. Once the €0.5 price band is actually implemented in 2017, the additional savings from the counterfactual reform decline sharply. The remaining savings after 2017 reflect the persistent difference in price levels generated by having introduced the narrower price band earlier.

A-6.2 Back-of-the-envelope calculation 2 - €0.1 price band size

We then consider a second back-of-the-envelope calculation in which the 2017 reform reduces the price band to €0.1 rather than €0.5. Figure A-18 shows that this further narrowing of the price band would have generated additional savings of up to €1.5 million per month, corresponding to approximately 3% of total pharmaceutical expenditure.

Figure A-18: Change in the total expenditure - €0.1 price band



Notes: Data source: Fimea (2009–2020).